All CCN Lab Reports

COURSE CODE: SE-305L

COURSE TITLE:

Computer Communications & Network

LAB TITLE(s):

Report Serial No.	Lab Title(s)
01	UTP Cable Configuration
02	Network Devices like, Switch, Router etc.
03	LAN Setup & Network Troubleshooting
04	DHCP Configuration
05	Two Router Network with Static Routing
06	VLAN Creation in Cisco Packet Tracer
07	Inter-VLAN Routing (Router-on-a-Stick)
08	IP Addresses & Subnetting
09	DNS Configuration
10	Wireless Network Setup

Submitted To: Engr. Riasat Khan.

Submitted By: Muhammad Danial.

Registration – No: 24ABSWE0021.

Semester: 4th

Date: 21-07-2026.

Dept. of Software Engineering

UET Peshawar, Abbottabad Campus.

Instructor Sign : ________________

Lab # 01

UTP Cable Configuration: Straight and Cross-Over

1. Objective

The objective of this lab was to understand the internal wiring standards of Unshielded Twisted Pair (UTP) cable and to practically prepare both Straight-Through and Cross-Over Ethernet cables. Specifically:

- Physical cable preparation
- Create a network using a Straight-Through cable.
- Create a network using a Cross-Over cable.
- Test connectivity between devices using the Ping command.

2. Equipment / Tools Used

- Cat5e/Cat6 UTP cable
- RJ-45 connectors
- Crimping tool
- Cable tester (LAN tester)
- Wire stripper / cutter
- Cisco Packet Tracer software

3. Pin Configuration (T568A vs T568B)

A UTP (Unshielded Twisted Pair) cable contains four twisted pairs of copper wires, each pair color-coded, terminated with RJ-45 connectors. The order in which the eight wires are arranged inside the connector follows one of two EIA/TIA standards: T568A and T568B. Depending on which standard is used at each end of the cable, the resulting cable behaves differently on the network:

A Straight-Through cable uses the same wiring standard (T568B) on both ends. It is used to connect dissimilar devices, such as a PC to a switch.

A Cross-Over cable uses T568A on one end and T568B on the other. The transmit and receive pairs are swapped, which allows two similar devices, such as two PCs, to communicate directly without an intermediate switch.

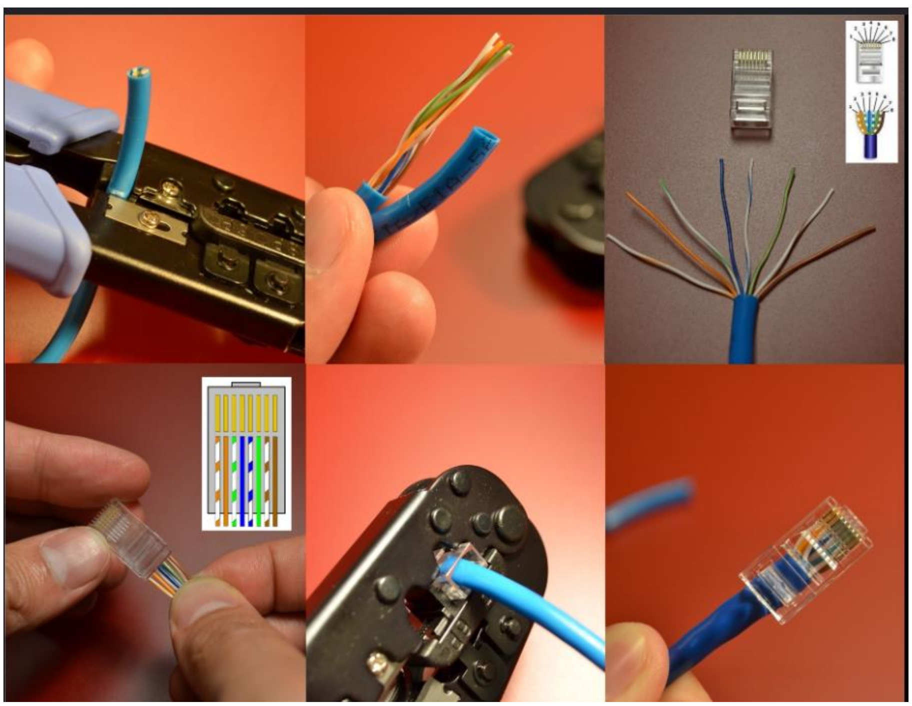
Pin	T568A Color	T568B Color	Pair
1	White/Green	White/Orange	Pair 2 / 3
2	Green	Orange	Pair 2 / 3
3	White/Orange	White/Green	Pair 3 / 2

Pin	T568A Color	T568B Color	Pair
4	Blue	Blue	Pair 1
5	White/Blue	White/Blue	Pair 1
6	Orange	Green	Pair 3 / 2
7	White/Brown	White/Brown	Pair 4
8	Brown	Brown	Pair 4

4. Physical Cable Preparation

Procedure

- The outer jacket of the UTP cable was stripped off using a cable stripper, exposing about 1.5 inches of the four twisted pairs.
- The pairs were untwisted and arranged according to the required standard (T568B for both ends of the straight cable, and T568A on one end of the cross-over cable).
- The wires were trimmed evenly and inserted into the RJ-45 connector in the correct pin order.
- The connector was crimped firmly using the crimping tool so that the metal pins pierced through each wire.
- The same steps were repeated for the second end, using the appropriate standard depending on the cable type being made.
- Each finished cable was checked using the cable tester to confirm that all eight pins showed continuity in the correct sequence.



Main Difference Between T568A and T568B

T568A	T568B
Green pair on pins 1 & 2	Orange pair on pins 1 & 2
Orange pair on pins 3 & 6	Green pair on pins 3 & 6

Cisco Packet Tracer Simulation

Task 1: Straight-Through Cable (PC ↔ Switch)

Step 1 – Add Devices: from the bottom-left device menu, two PCs were added from End Devices, and one 2960 Switch was added from Network Devices → Switches, forming the topology PC0 – Switch0 – PC1.

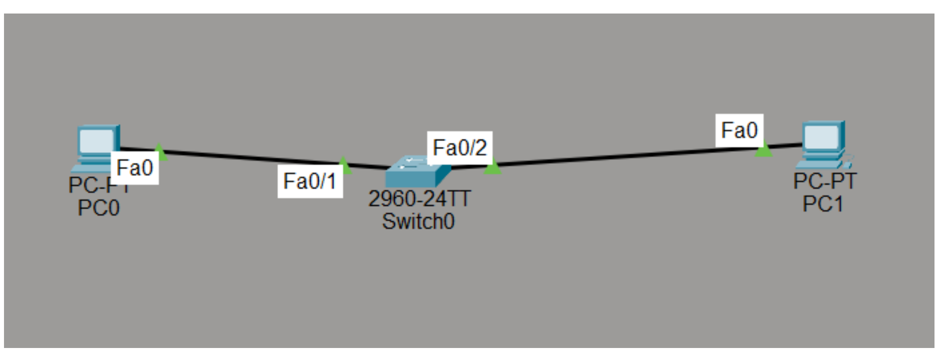
Step 2 – Select the Cable: the Connections tool (lightning-bolt icon) was opened and Copper Straight-Through was selected.

Step 3 – Connect Devices: PC0 FastEthernet0 was connected to Switch FastEthernet0/1, and PC1 FastEthernet0 was connected to Switch FastEthernet0/2. After a few seconds the link LEDs turned green, indicating an active connection.

Step 4 – Configure IP Addresses: using Desktop → IP Configuration on each PC, the following addresses were assigned (gateway left empty since no router is involved):

PC0	IP: 192.168.1.1 | Subnet Mask: 255.255.255.0
PC1	IP: 192.168.1.2 | Subnet Mask: 255.255.255.0

Step 5 – Test Connectivity: from PC0's Command Prompt, the command "ping 192.168.1.2" was executed. The expected result is a successful reply from PC1 with 4 packets sent, 4 received, and 0 lost, confirming the network is working correctly.



```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.2

Pinging 192.168.1.2 with 32 bytes of data:

Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

Task 2: Cross-Over Cable (PC ↔ PC)

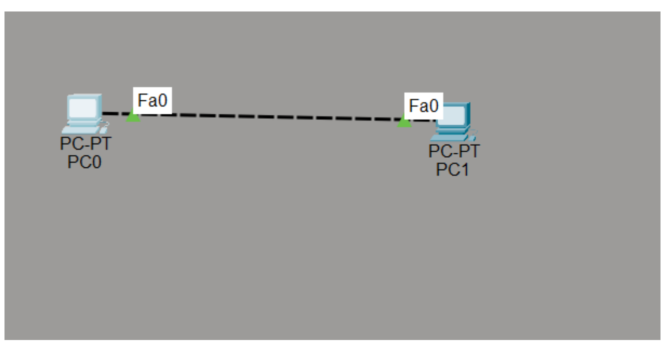
Step 1: the switch was removed from the workspace, leaving only two PCs connected directly as PC0 – PC1.

Step 2: the cable type was changed to Copper Cross-Over, and PC0 FastEthernet0 was connected directly to PC1 FastEthernet0.

Step 3: new IP addresses were assigned on a separate subnet:

PC0	IP: 192.168.10.1 | Subnet Mask: 255.255.255.0
PC1	IP: 192.168.10.2 | Subnet Mask: 255.255.255.0

Step 4: from PC0, the command "ping 192.168.10.2" was executed. Successful replies confirmed that the direct PC-to-PC connection was working correctly over the cross-over cable.



```
C:\>ping 192.168.10.2

Pinging 192.168.10.2 with 32 bytes of data:

Reply from 192.168.10.2: bytes=32 time<1ms TTL=128
Reply from 192.168.10.2: bytes=32 time<1ms TTL=128
Reply from 192.168.10.2: bytes=32 time<1ms TTL=128
Reply from 192.168.10.2: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.10.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

5.Auto-Connect Option

Packet Tracer also provides an “Automatically Choose Connection Type” option (lightning bolt with a question mark), which automatically selects the correct cable, straight-through or cross-over, based on the two devices being connected. This is convenient for quick simulations, but it can hide the wiring logic that this lab is meant to teach, so the specific cable types were selected manually for both experiments.

6. Cable Selection Reference

Devices Connected	Cable Used
PC ↔ Switch	Straight-Through
PC ↔ Hub	Straight-Through
Router ↔ Switch	Straight-Through
PC ↔ PC	Cross-Over
Switch ↔ Switch	Cross-Over
Router ↔ Router	Cross-Over

6. Observations

When Packet Tracer automatically detects the connection, it displays a link light at each end of the cable: a green dot indicates an active link, while a red or blinking dot indicates a wiring or configuration issue.

- Connecting two PCs directly with a straight-through cable resulted in no link, since both devices transmit and receive on the same pins.
- Replacing it with a cross-over cable established a successful link between the two PCs, and the ping command returned successful replies.
- The PC-to-switch connection worked correctly with a straight-through cable, and failed to establish a link when a cross-over cable was used instead.

- The physically crimped cables passed the continuity test on the cable tester, with all eight LEDs lighting up in the correct 1-8 sequence for the straight cable, and in the corresponding swapped sequence for the cross-over cable.

7. Conclusion

This lab provided practical understanding of the two most common UTP wiring standards, T568A and T568B, and how their combination determines whether a cable is Straight-Through or Cross-Over. Preparing the cables manually with a crimping tool and verifying them with a cable tester reinforced the theoretical pinout, while the Cisco Packet Tracer simulation confirmed the logical rule that similar devices require a cross-over cable and dissimilar devices require a straight-through cable. Overall, the lab connected the physical hardware process with its logical networking behavior.

Lab # 02

Network devices like switch, hub, router, bridge, repeater

Introduction:

Networking and internetworking devices are used to connect computers and enable communication within and between networks. Devices such as Hub, Switch, Bridge, Repeater, and Router perform different functions to ensure efficient data transmission. This lab demonstrates the basic operation and purpose of these devices using Cisco Packet Tracer.

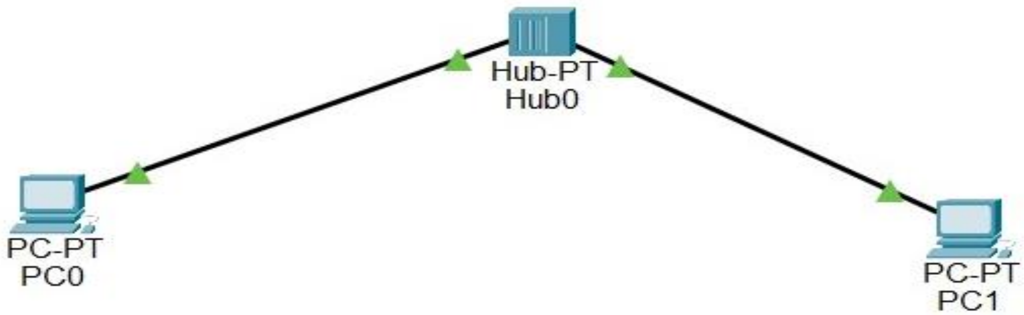
Lab Task:

To study and demonstrate the functions of various networking and internetworking devices, including the **Switch, Hub, Router, Bridge, and Repeater**, using Cisco Packet Tracer.

HUB:

Procedure:

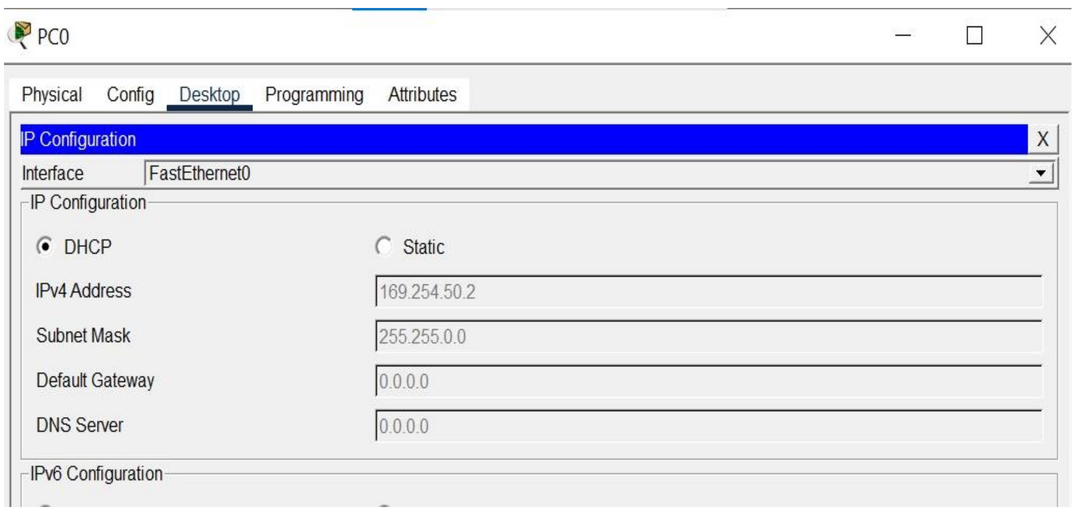
1. Open Cisco Packet Tracer and create a new project.
2. Drag and place one Hub (Hub-PT) from Network Devices → Hubs onto the workspace.
3. Drag and place two PCs (PC0 and PC1) from End Devices.
4. Select Copper Straight-Through Cable from the Connections menu.
5. Connect PC0 (FastEthernet0) to Hub Port 0.
6. Connect PC1 (FastEthernet0) to Hub Port 1.
7. Wait until the link lights turn green, indicating that the physical connections are active.



8. Configure the IP address of PC0:

    IP Address: 192.168.1.1

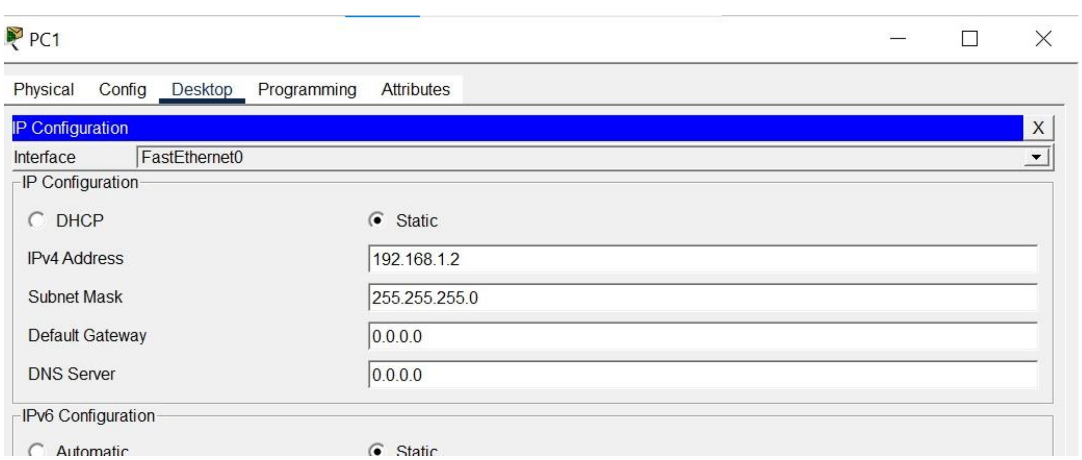
    Subnet Mask: 255.255.255.0



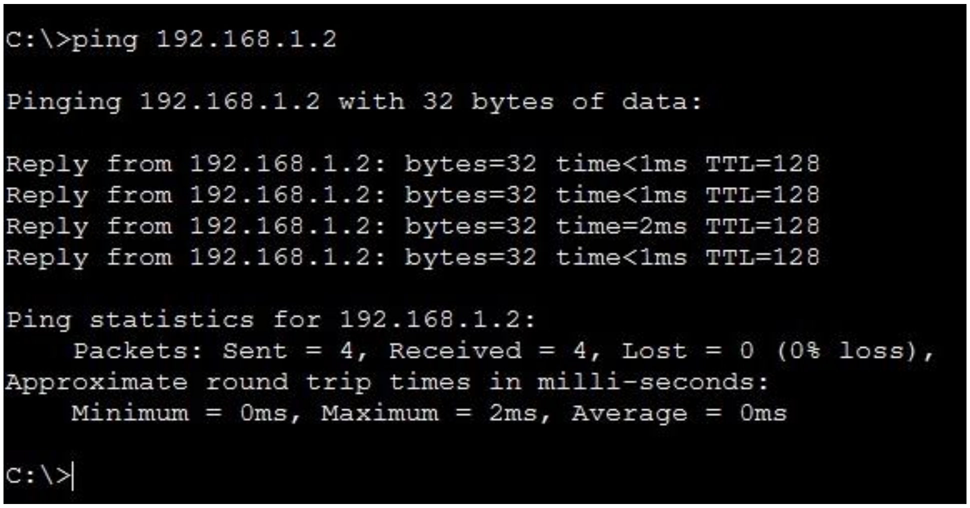
9. Configure the IP address of PC1:

    IP Address: 192.168.1.2

    Subnet Mask: 255.255.255.0



10. Open Desktop → Command Prompt on PC0.
11. Execute the command:
12. ping 192.168.1.2



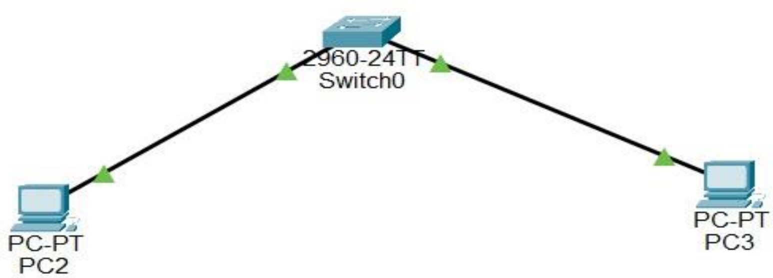
```
C:\>ping 192.168.1.2

Pinging 192.168.1.2 with 32 bytes of data:

Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time=2ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 2ms, Average = 0ms

C:\>
```

13. Observe that all packets are successfully received, confirming communication between the two PCs through the Hub.

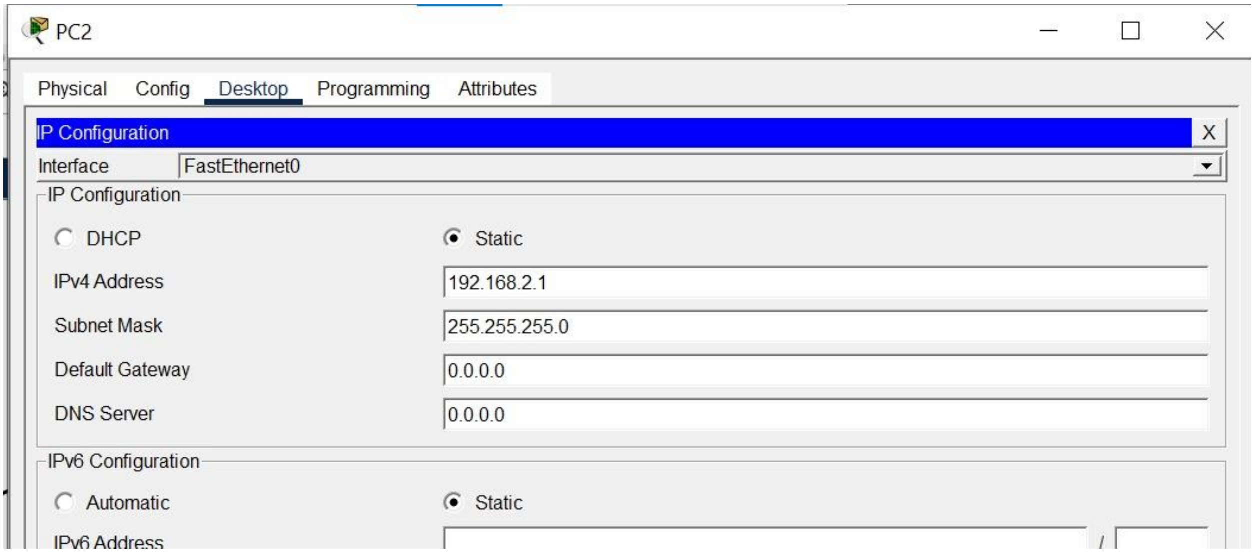
SWITCH:

Procedure:

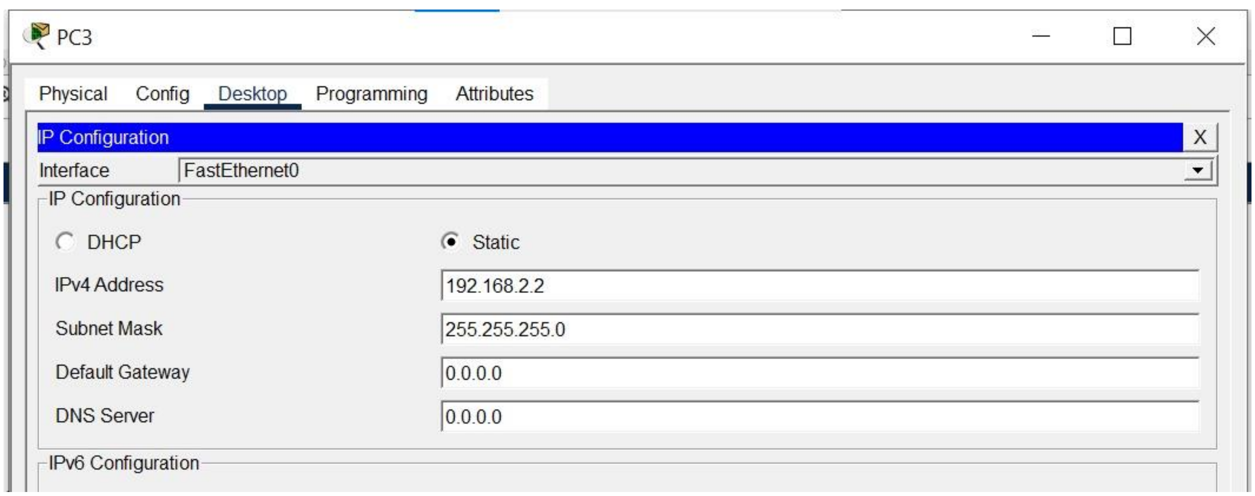
1. Open Cisco Packet Tracer and create a new project.
2. Place one Switch (2960) and two PCs (PC2 and PC3) on the workspace.
3. Connect both PCs to the Switch using Copper Straight-Through cables.
4. Verify that all switch ports are active and the link lights are green.



5. Configure the IP address of PC2:
   - IP Address: 192.168.2.1
   - Subnet Mask: 255.255.255.0

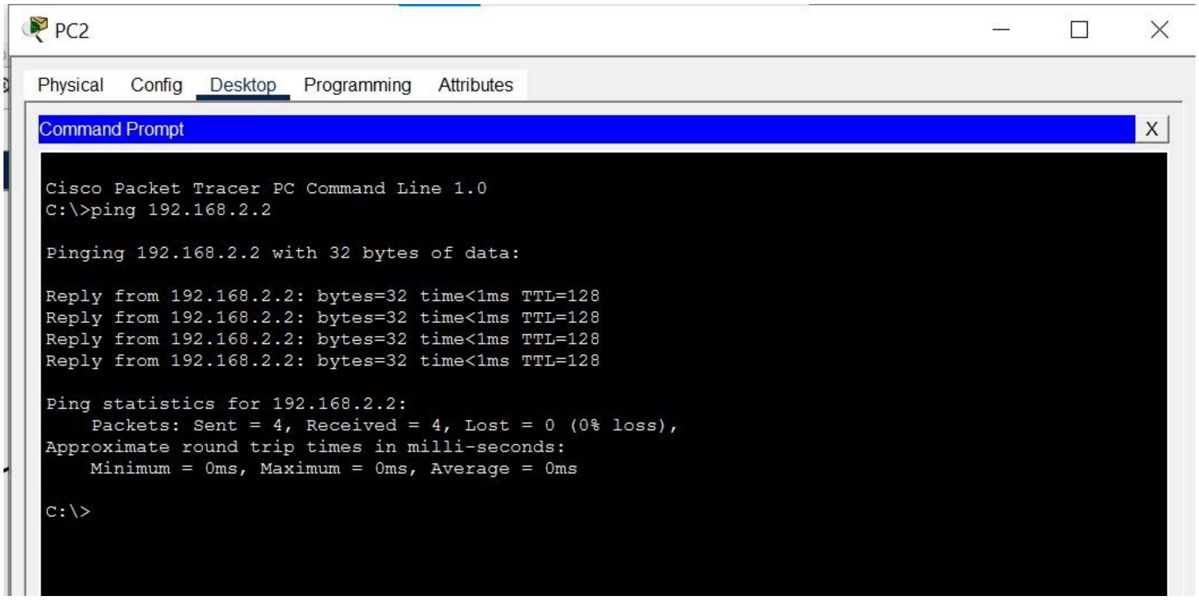


6. Configure the IP address of PC3:
    - IP Address: 192.168.2.2
    - Subnet Mask: 255.255.255.0



7. Open Command Prompt on PC2.
8. Execute the following command:

ping 192.168.2.2



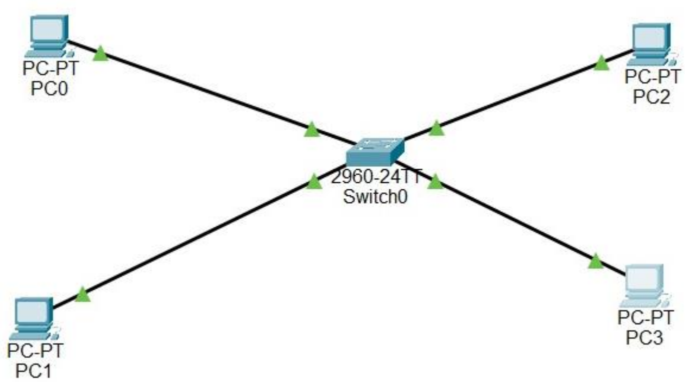

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.2.2

Pinging 192.168.2.2 with 32 bytes of data:

Reply from 192.168.2.2: bytes=32 time<1ms TTL=128
Reply from 192.168.2.2: bytes=32 time<1ms TTL=128
Reply from 192.168.2.2: bytes=32 time<1ms TTL=128
Reply from 192.168.2.2: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.2.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

9. Observe the ping results to verify successful communication between the two PCs through the Switch

BRIDGE:

Procedure:

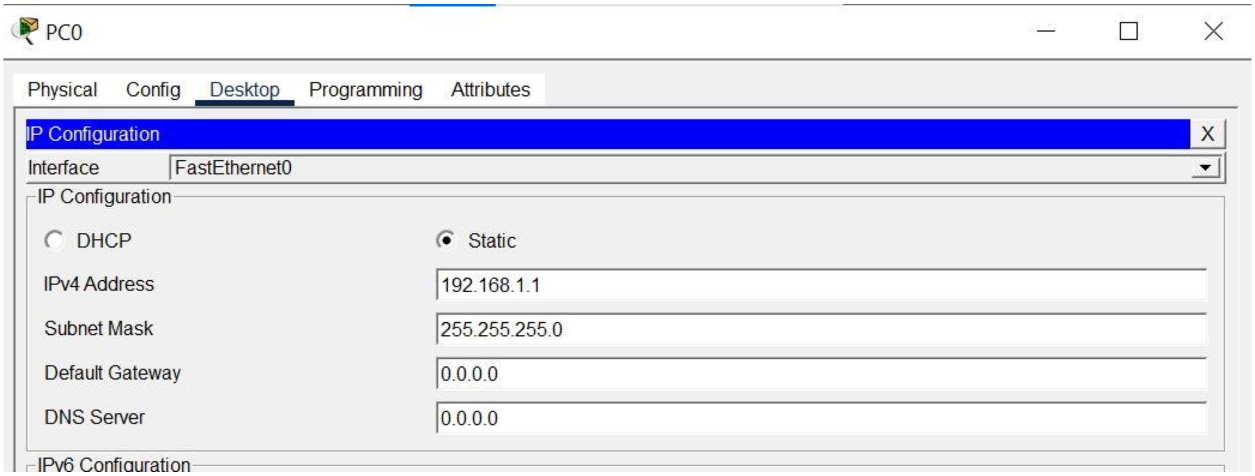
1. Opened Cisco Packet Tracer.
2. Added one Cisco 2960 switch to represent a bridge.
3. Added four PCs representing two LAN segments.
4. Connected all PCs to the switch using Copper Straight-Through cables.



5. Configure the IP address of PC0:

   IP Address: 192.168.1.1

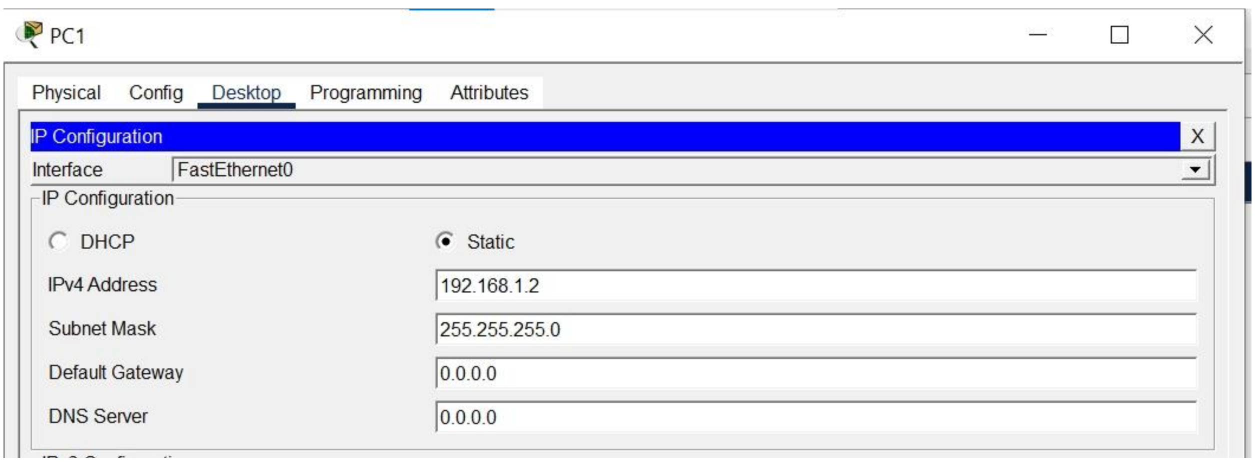
   Subnet Mask: 255.255.255.0



6. Configure the IP address of PC1:

   IP Address: 192.168.1.2

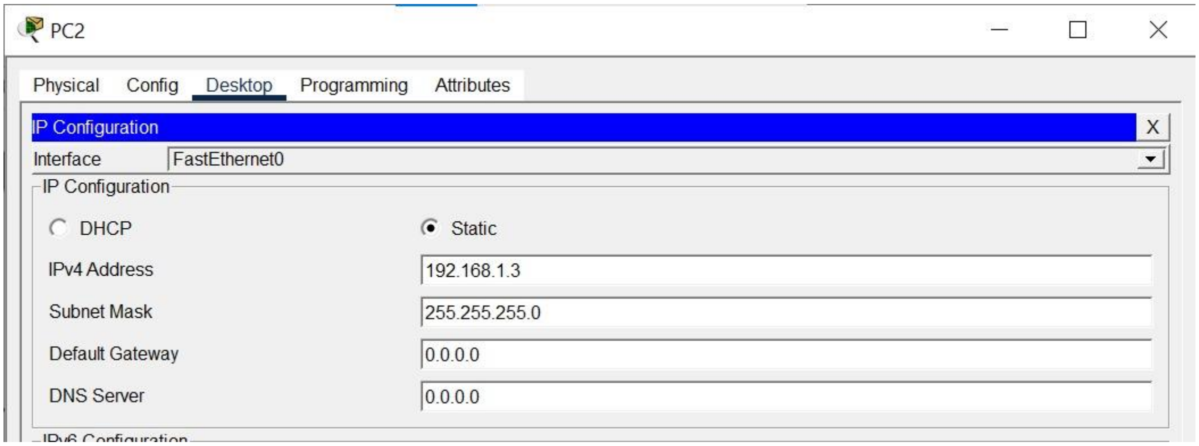
   Subnet Mask: 255.255.255.0



7. Configure the IP address of PC2:

   IP Address: 192.168.1.3

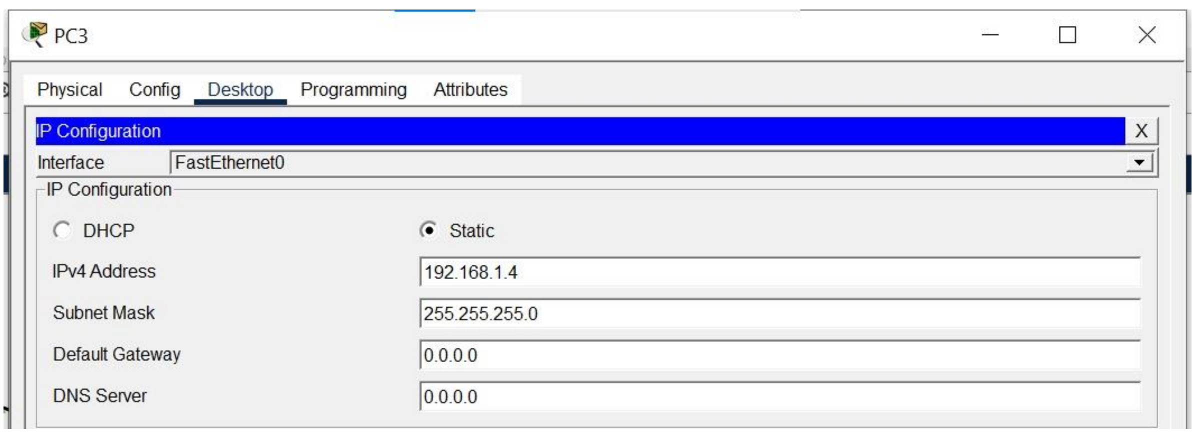
   Subnet Mask: 255.255.255.0



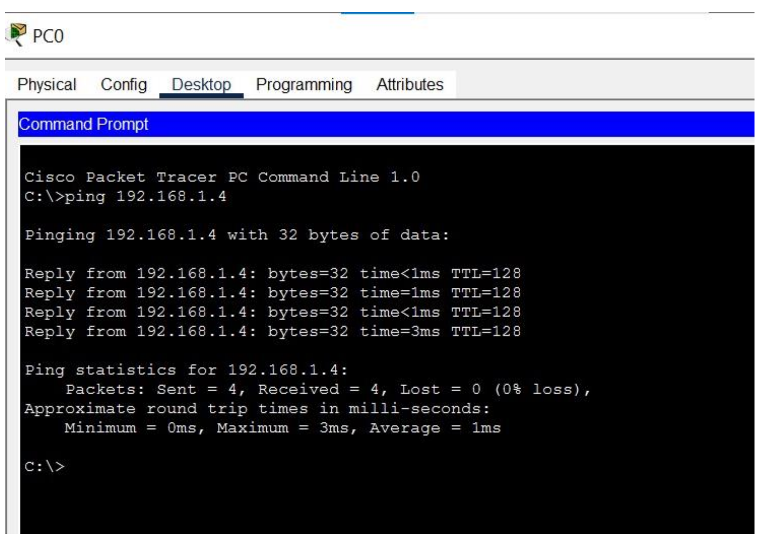
8. Configure the IP address of PC3:

   IP Address: 192.168.1.4

   Subnet Mask: 255.255.255.0



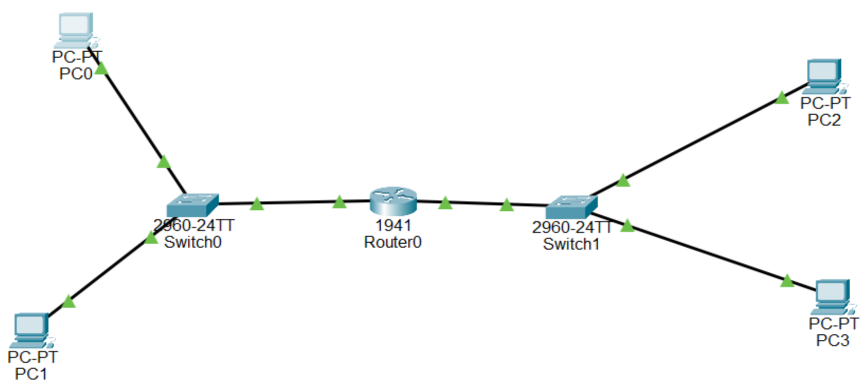
9. Verified communication using the ping command



ROUTER:

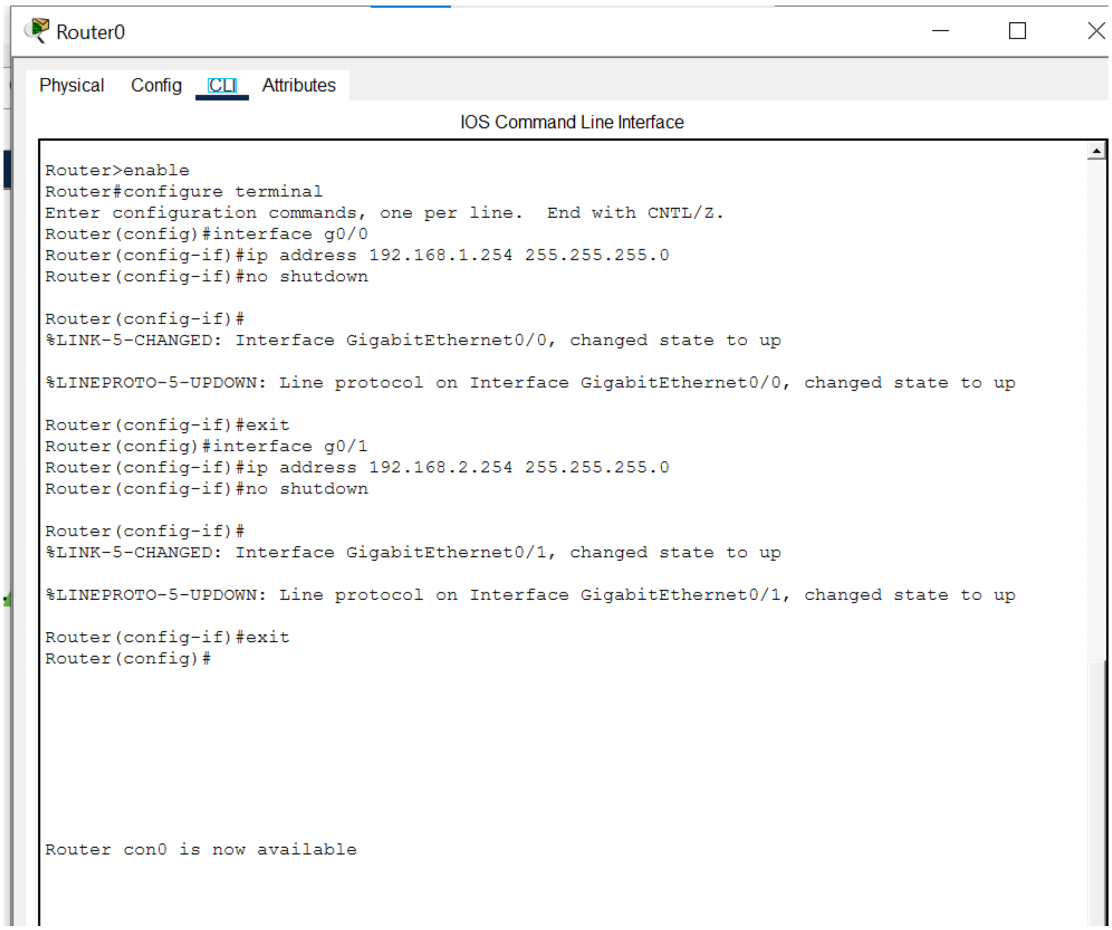
Procedure:

1. Open Cisco Packet Tracer and create a new project.
2. Place one Router (1941), two Switches (2960), and four PCs on the workspace.
3. Connect PC0 and PC1 to Switch1 using Copper Straight-Through cables.
4. Connect Switch1 to the GigabitEthernet0/0 interface of the router.
5. Connect the GigabitEthernet0/1 interface of the router to Switch2.
6. Connect PC2 and PC3 to Switch2 using Copper Straight-Through cables.
7. Verify that all links are active (green).

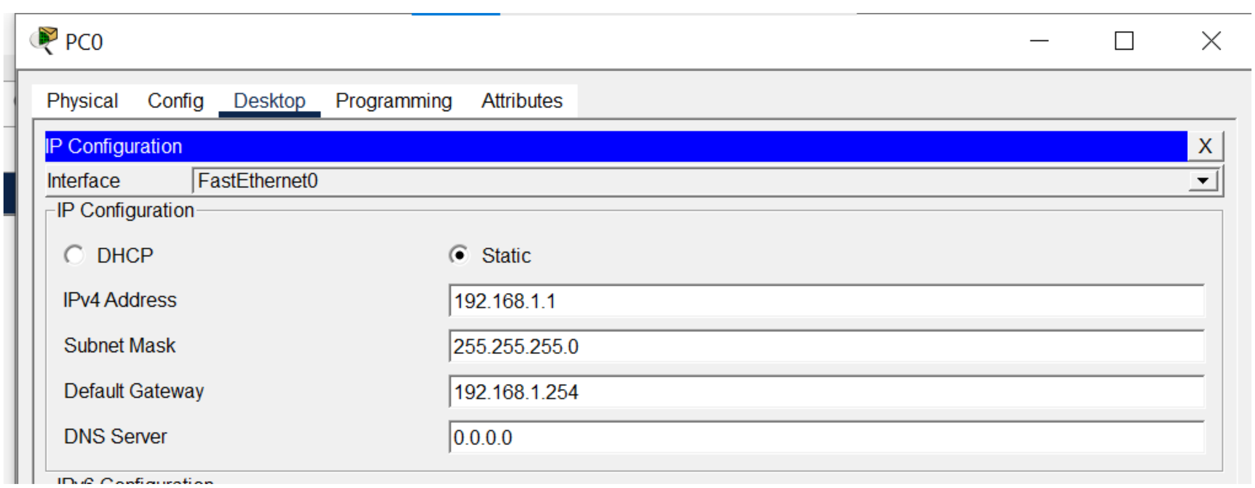


8. Configure the router interfaces with the following IP addresses:
   a. GigabitEthernet0/0: 192.168.1.254/24
   b. GigabitEthernet0/1: 192.168.2.254/24

   Enable both router interfaces using the no shutdown command.

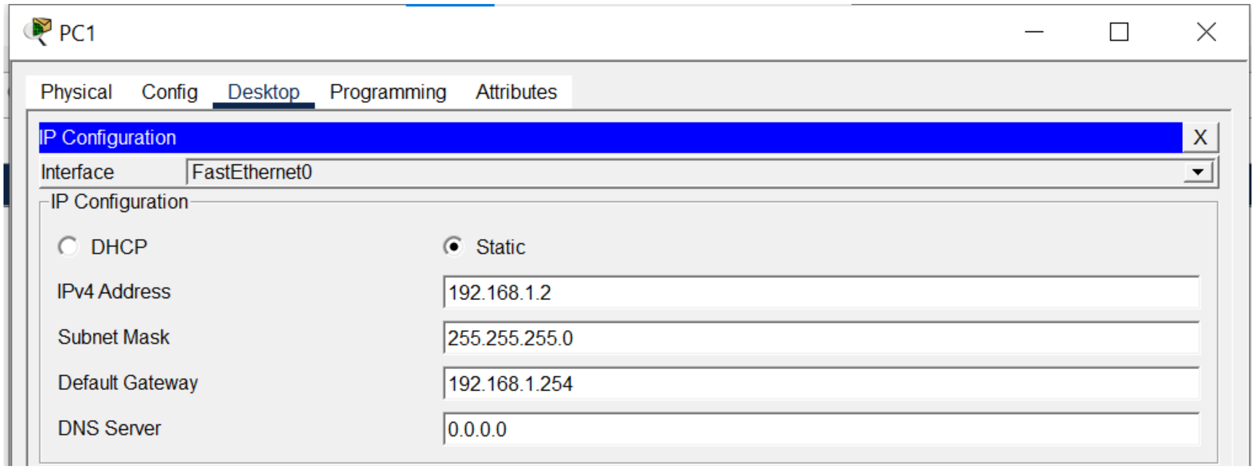


9. Configure the PC0 with the following network settings:
- IP Address 192.168.1.1
- Subnet Mask 255.255.255.0
- Default Gateway 192.168.1.254



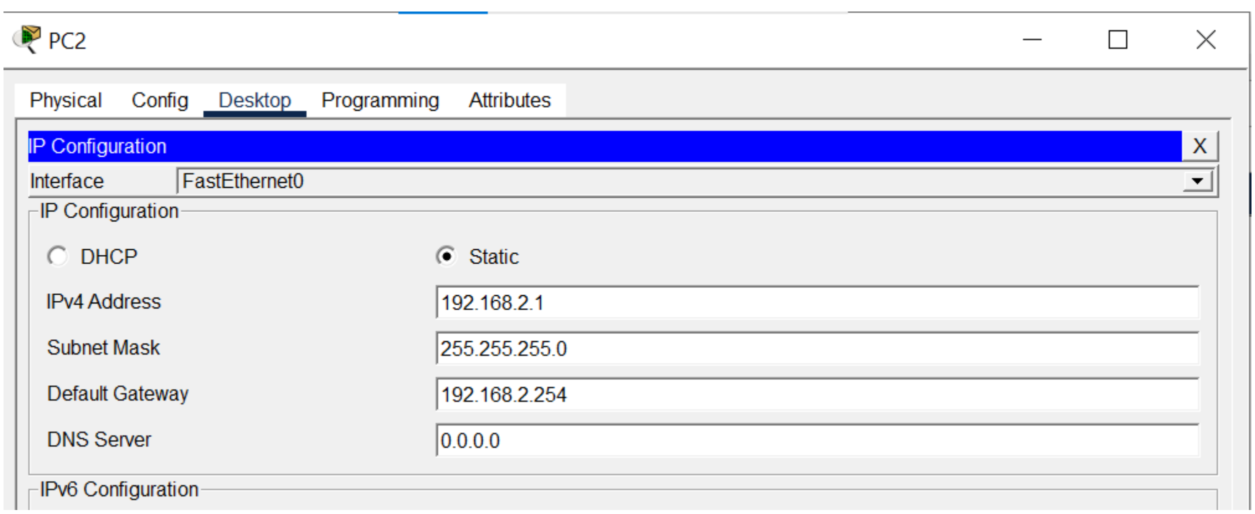
10. Configure the PC1 with the following network settings:

- IP Address 192.168.1.2
- Subnet Mask 255.255.255.0
- Default Gateway 192.168.1.254



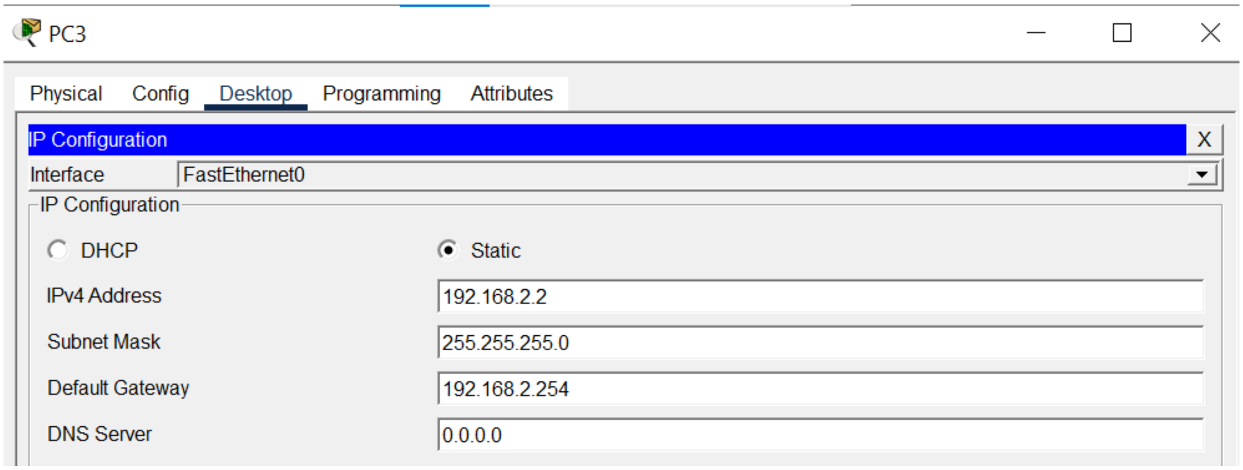
11. Configure the PC2 with the following network settings:

- IP Address 192.168.2.1
- Subnet Mask 255.255.255.0
- Default Gateway 192.168.2.254



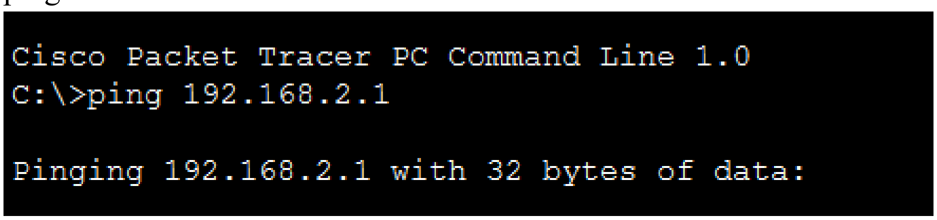
12. Configure the PCs with the following network settings:

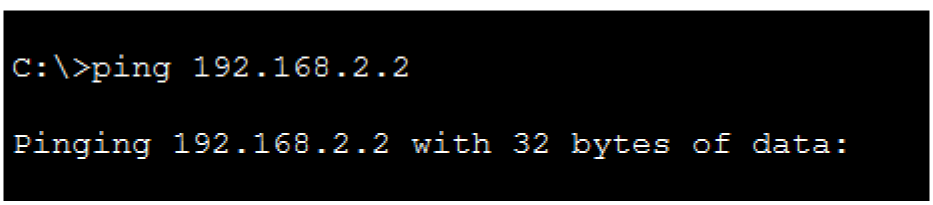
- IP Address 192.168.2.2
- Subnet Mask 255.255.255.0
- Default Gateway 192.168.2.254



13. Test connectivity by executing the following commands from PC0:

- ping 192.168.2.1
- ping 192.168.2.2





14. Observe the successful ping replies, confirming communication between the two different networks through the router.

<u>REPEATER:</u>

Procedure:

1. Opened Cisco Packet Tracer and created a new project.
2. Searched for the Repeater device under Network Devices. However, the installed version of Cisco Packet Tracer did not include the Repeater device, as it has been removed from newer versions.
3. In versions of Cisco Packet Tracer where the Repeater is available, the following steps are performed:

6. Add one Repeater and two PCs to the workspace.
2. Connect PC0 to the Repeater using a Copper Straight-Through cable.
3. Connect the Repeater to PC1 using another Copper Straight-Through cable.
4. Configure the IP addresses of both PCs so they belong to the same network.
5. Verify the connections between the devices.
6. Open the Command Prompt on PC0 and use the ping command to test connectivity with PC1.
7. Switch to Simulation Mode and observe the packet passing through the Repeater, confirming successful communication.

Lab # 03

Basic LAN Setup, Network Troubleshooting, and Device Security Configuration

Objectives:

- To create a Local Area Network (LAN) using a switch
- To assign static IP addresses to PCs
- To test connectivity using ping command
- To perform basic network troubleshooting
- To configure hostname and passwords on network devices

Tools / Requirements:

- Cisco Packet Tracer
- 1 Switch (2960)
- 2–3 PCs
- Copper Straight-Through Cables

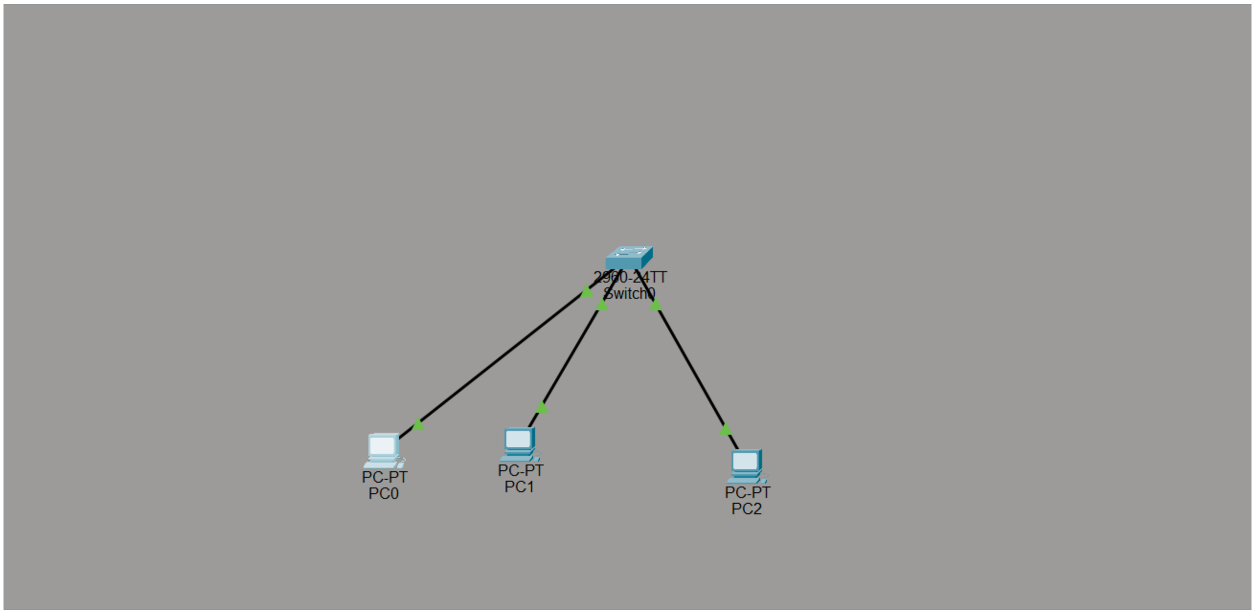
Part# 01

Step 1: Add Devices

- Added:
  - 1 Switch (2960)
  - 3 PCs (PC0, PC1, PC2)

Step 2: Connect Devices

- Used **Copper Straight-Through Cable**
- Connections:
    - PC0 → Switch (Fa0/1)
    - PC1 → Switch (Fa0/2)
    - PC2 → Switch (Fa0/3)



Step 3: Assign IP Addresses

PC	IP Address	Subnet Mask
PC0	192.168.1.1	255.255.255.0
PC1	192.168.1.2	255.255.255.0
PC2	192.168.1.3	255.255.255.0

Step 4: Test Connectivity

From PC0:

ping 192.168.1.2
ping 192.168.1.3

From PC1:

ping 192.168.1.1
ping 192.168.1.3

From PC2:

ping 192.168.1.1
ping 192.168.1.2

Successfully received replies:

```
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>PING 192.168.1.2

Pinging 192.168.1.2 with 32 bytes of data:

Request timed out.
Request timed out.
```

Top

Part # 02

Step 1: Introduced Errors

- Changed IP address to wrong network (192.168.2.x)
- Shutdown switch interface
- Used incorrect cable

Step 2: Identifying Issues

- Used ping → Request timed out
- Checked IP configuration on PCs
- Used switch command:

show ip interface brief

- Observed interface status (down)

```
Switch>ENABLE
Switch#SHOW IP ADDRESS BRIEFLY
                ^
% Invalid input detected at '^' marker.

Switch#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
FastEthernet0/1        unassigned      YES manual up                    up
FastEthernet0/2        unassigned      YES manual up                    up
FastEthernet0/3        unassigned      YES manual up                    up
FastEthernet0/4        unassigned      YES manual down                  down
FastEthernet0/5        unassigned      YES manual down                  down
FastEthernet0/6        unassigned      YES manual down                  down
FastEthernet0/7        unassigned      YES manual down                  down
FastEthernet0/8        unassigned      YES manual down                  down
FastEthernet0/9        unassigned      YES manual down                  down
FastEthernet0/10       unassigned      YES manual down                  down
FastEthernet0/11       unassigned      YES manual down                  down
FastEthernet0/12       unassigned      YES manual down                  down
FastEthernet0/13       unassigned      YES manual down                  down
FastEthernet0/14       unassigned      YES manual down                  down
FastEthernet0/15       unassigned      YES manual down                  down
FastEthernet0/16       unassigned      YES manual down                  down
FastEthernet0/17       unassigned      YES manual down                  down
FastEthernet0/18       unassigned      YES manual down                  down
FastEthernet0/19       unassigned      YES manual down                  down
FastEthernet0/20       unassigned      YES manual down                  down
FastEthernet0/21       unassigned      YES manual down                  down
 --More--
```

Step 3: Fixing Issues

- Corrected IP address
- Enabled interface:

no shutdown

- Replaced with correct cable

```
C:\>ping 192.168.1.2

Pinging 192.168.1.2 with 32 bytes of data:

Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time=1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time=1ms TTL=128

Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>
```

Part 3: Hostname & Password Configuration

Step 1: Set Hostname

enable
configure terminal
hostname LabSwitch

Step 2: Set Console Password

line console 0
password cisco
login

Step 3: Set Enable Secret

enable secret class

Step 4: Encrypt Passwords

service password-encryption

Step 5: Save Configuration

end
copy running-config startup-config

```
Vlan1                  unassigned      YES manual administratively down down
Switch#
Switch#enable
Switch#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#hostname labswitch
labswitch(config)#line console 0
labswitch(config-line)#password cisco
labswitch(config-line)#login
labswitch(config-line)#exit
labswitch(config)#enable secret class
labswitch(config)#service password-encrytion
                                        ^
% Invalid input detected at '^' marker.

labswitch(config)#service password-encryption
labswitch(config)#end lab memory
                      ^
% Invalid input detected at '^' marker.

labswitch(config)#end
labswitch#
%SYS-5-CONFIG_I: Configured from console by console
write memory
Building configuration...
[OK]
labswitch#
```

Lab # 04

DHCP Configuration (Switch Based Network & Router)

Description of Lab:

This lab demonstrates how to configure a router as a DHCP server so that PCs connected via a switch on the same LAN automatically receive IP addressing, while reserving a specific range of addresses for static/manual use.

Devices Used:

- 1 × Router acting as DHCP server
- 1 × Switch, connecting PCs to the router's LAN interface
- 3 × PCs configured for automatic (DHCP) IP addressing.
- Connecting cables (Copper Straight-Through)

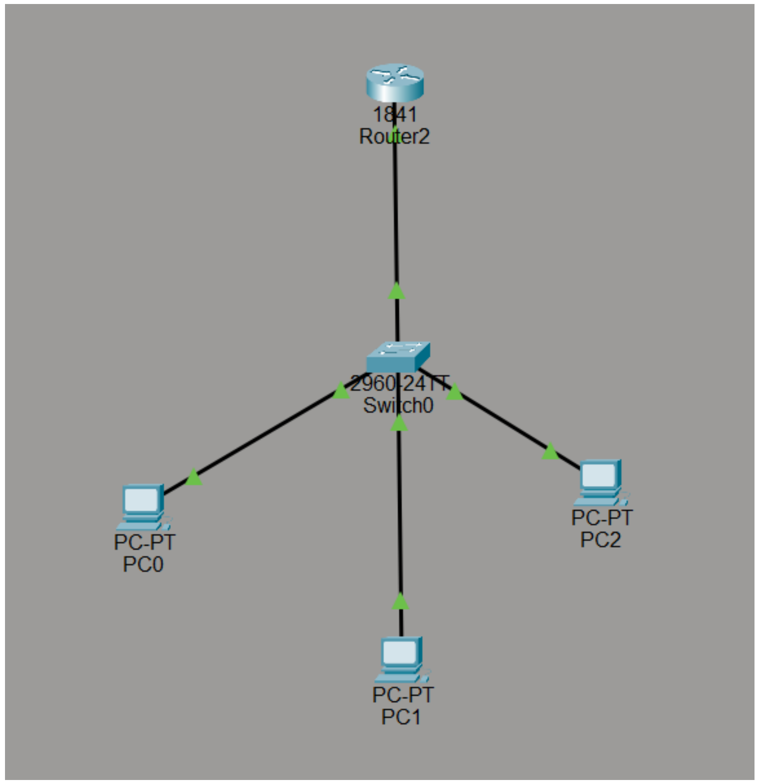


FIGURE 1

Configuration Steps:

1. Accessed Router2 via CLI tab and entered privileged/global config mode.
2. Excluded the address range reserved for static devices (192.168.1.1–192.168.1.10) done before pool creation to prevent conflicts.
3. Created a DHCP pool named LAN-POOL.
4. Defined the network and subnet mask inside the pool.
5. Set the default gateway (router's LAN interface IP).
6. Set a DNS server address "8.8.8.8".
7. Configured PCs to obtain IP addressing via DHCP (Desktop → IP Configuration).
8. Verified lease assignment using "show ip dhcp binding".

IP Addressing Table:

Device Name	Device Type	IP Address	Subnet Mask	Default Gateway	DNS Server
Router	DHCP Server	192.168.1.1	255.255.255.0	–	–
PC0	Client PC	192.168.1.11	255.255.255.0	192.168.1.1	8.8.8.8
PC1	Client PC	192.168.1.12	255.255.255.0	192.168.1.1	8.8.8.8
PC2	Client PC	192.168.1.13	255.255.255.0	192.168.1.1	8.8.8.8

CLI Commands:

Router> enable

Router# configure t

Router(config)# interface fa0/0

Router(config-if)#ip address 192.168.1.1 255.25.255.0

Router(config-if)#no shutdown

Router(config-if)#exit

Router(config)# ip dhcp excluded-address 192.168.1.1 192.168.1.10

Router(config)# ip dhcp pool LAN-POOL

Router(dhcp-config)# network 192.168.1.0 255.255.255.0

Router(dhcp-config)# default-router 192.168.1.1

Router(dhcp-config)# dns-server 8.8.8.8

Router(dhcp-config)# exit

Router# show ip dhcp binding

Simulation Results:

```
Router>enable
Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface fast ethernet 0/0
                                        ^
% Invalid input detected at '^' marker.

Router(config)#interface fa0/0
Router(config-if)#ip address 192.168.1.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-3-UPDOWN: Interface FastEthernet0/0, changed state to down

Router(config-if)#exit
Router(config)#ip dhcp excluded-address 192.168.1.1 192.168.1.10
Router(config)#ip dhcp pool LAN-POOL
Router(dhcp-config)#192.168.1.0 255.255.255.0
                    ^
% Invalid input detected at '^' marker.

Router(dhcp-config)#network 192.168.1.0 255.255.255.0
Router(dhcp-config)#default router 192.168.1.1
                            ^
% Invalid input detected at '^' marker.

Router(dhcp-config)#default-router 192.168.1.1
Router(dhcp-config)#dns-server 8.8.8.8
Router(dhcp-config)#exit
Router(config)#exit
```

Figure 2:-Router IP Configuration.

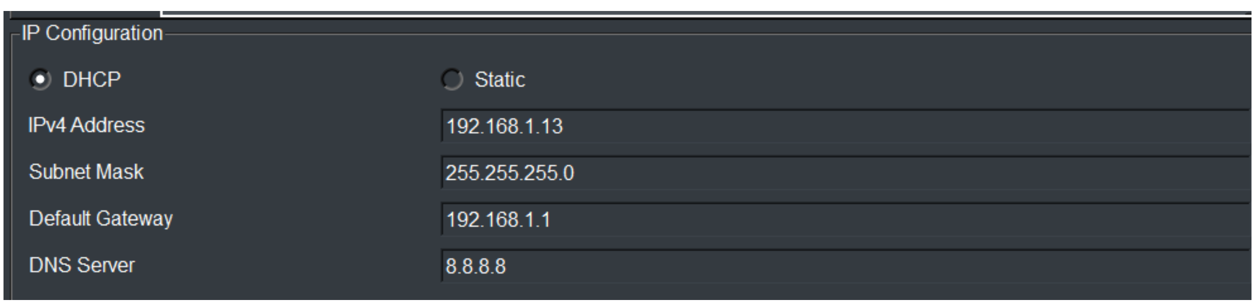


Figure 3:-PC0 IP Configuration.

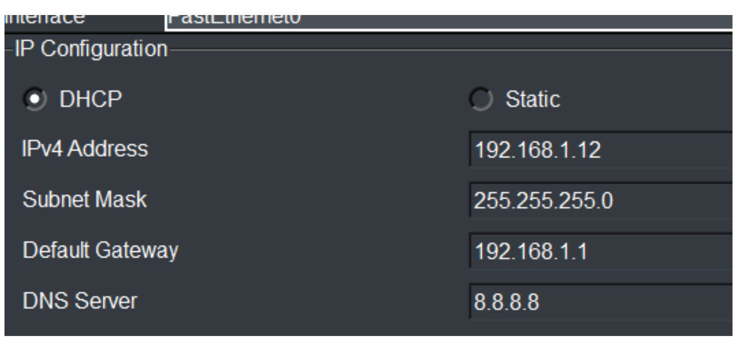


Figure 4: PC1 IP Configuration.

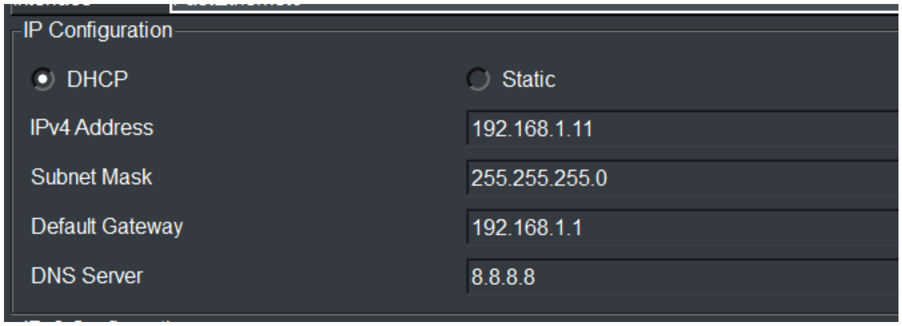


Figure 5: PC2 IP Configuration.

Conclusion:

The router was successfully configured as a DHCP server, and PCs connected through a switch on the same subnet received automatic IP addressing without requiring DHCP relay configuration, since no Layer 3 boundary separated the server from the clients. The exclusion range was respected, confirming correct order of configuration (exclusion set before pool creation). This demonstrates that DHCP relay (ip helper-address) is only necessary when the server and clients reside on different subnets connected via a router, not in a flat switch-based LAN.

Lab # 05

Two Router Network with Static Routing

Two-Router Network with Static Routing:

Connect two routers, each with its own LAN segment (a switch and a few PCs). Configure static routes on both routers to enable communication between the two distinct LANs

---

Objective

- To configure a network using two routers.
- To connect two separate LANs using static routing.
- To verify communication between devices on different networks.

Equipment Required

- 2 Routers

- 2 Switches
- 4 PCs (or more)
- UTP Straight-Through and Cross-Over Cables (or Auto cable in Packet Tracer)
- Cisco Packet Tracer

Theory

Static routing is a routing method in which routes are manually configured by the network administrator. It enables communication between different networks by specifying the path that packets should follow. In this experiment, two routers connect two separate LANs, allowing devices in both networks to communicate.

Procedure

1. Connect each router to a switch.
2. Connect PCs to their respective switches.
3. Connect the two routers together.
4. Assign IP addresses to all router interfaces and PCs.
5. Configure static routes on both routers.
6. Verify connectivity using the **ping** command.

Observation

- Static routes were configured successfully on both routers.
- PCs in different LANs were able to communicate.
- Ping requests received successful replies.

Result

Two LANs were successfully connected using static routing, and communication between devices on different networks was achieved.

Precautions

- Assign the correct IP addresses and subnet masks.
- Configure static routes accurately on both routers.
- Ensure all router interfaces are enabled (`no shutdown`).
- Verify cable connections before testing.

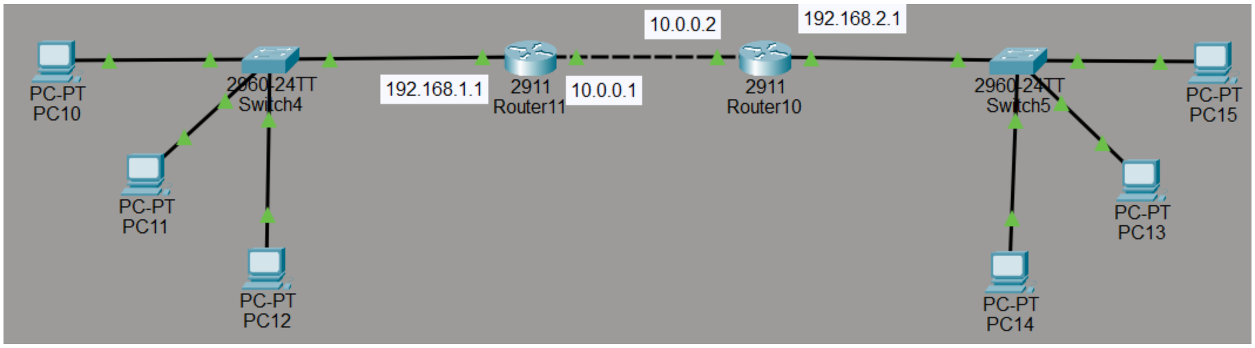


Figure: ***Topology***

```
Router>en
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface giga
Router(config)#interface gigabitEthernet 0/0
Router(config-if)#ip address 192.168.1.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-3-UPDOWN: Interface GigabitEthernet0/0, changed state to down

Router(config-if)#exit
Router(config)#interface gig
Router(config)#interface gigabitEthernet 0/1
Router(config-if)#ip address 10.0.0.1 255.255.255.252
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

Router(config-if)#exit
```

```
Router(config)#
Router(config)#
Router(config)#ip route 192.168.2.0 255.255.255.0 10.0.0.2
Router(config)#exit
Router#
```

Figure: ***Router 1 Configuration***

```
Router(config)#interface gig
Router(config)#interface gigabitEthernet 0/0
Router(config-if)#ip address 192.168.2.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-3-UPDOWN: Interface GigabitEthernet0/0, changed state to down

Router(config-if)#exit
Router(config)#
Router(config)#interface gig
Router(config)#interface gigabitEthernet 0/1
Router(config-if)#ip addr
Router(config-if)#ip address 10.0.0.2 255.255.255.252
Router(config-if)#no shut
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

Router(config-if)#exit
```

```
Router(config)#
Router(config)#
Router(config)#ip route 192.168.1.0 255.255.255.0 10.0.0.1
Router(config)#exit
Router#
```

Figure: ***Router 2 Configuration***

```
C:\>ping 192.168.2.3

Pinging 192.168.2.3 with 32 bytes of data:

Reply from 192.168.2.3: bytes=32 time<1ms TTL=126
Reply from 192.168.2.3: bytes=32 time<1ms TTL=126
Reply from 192.168.2.3: bytes=32 time=1ms TTL=126
Reply from 192.168.2.3: bytes=32 time<1ms TTL=126

Ping statistics for 192.168.2.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

Figure: ***Ping Verification From other LAN***

```
C:\>ping 192.168.1.3

Pinging 192.168.1.3 with 32 bytes of data:

Reply from 192.168.1.3: bytes=32 time=3ms TTL=128
Reply from 192.168.1.3: bytes=32 time=6ms TTL=128
Reply from 192.168.1.3: bytes=32 time=5ms TTL=128
Reply from 192.168.1.3: bytes=32 time=7ms TTL=128

Ping statistics for 192.168.1.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 3ms, Maximum = 7ms, Average = 5ms
```

Figure: ***Ping Verification from Self LAN***

Lab # 06

VLAN Creation Using Cisco Packet Tracer

LAB OBJECTIVES

- To create multiple VLANs on a Cisco switch.
- To assign switch ports to different VLANs.
- To connect PCs to the assigned ports.
- To verify communication within the same VLAN.
- To demonstrate that devices in different VLANs cannot communicate without a router.

REQUIREMENTS:

- Cisco Packet Tracer
- 1 Cisco 2960 Switch
- 4 PCs
- Copper Straight-Through Cables

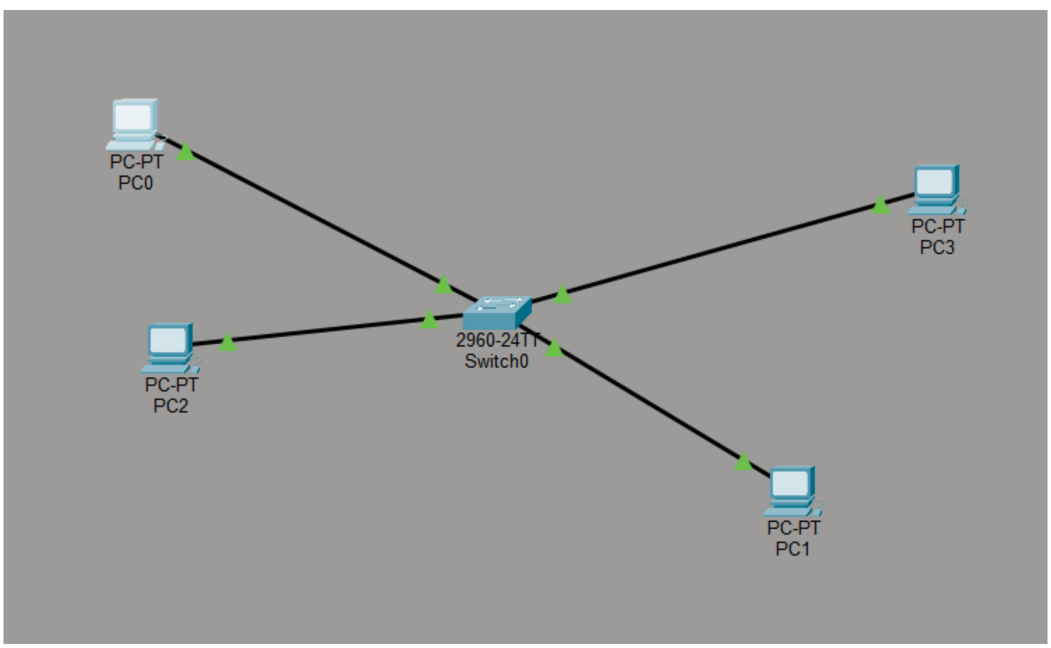
LAB TASKS

On a single switch, create two or more VLANs and assign specific switch ports to each VLAN. Connect PCs to these ports and demonstrate that devices in different VLANs cannot communicate without a router.

PROCEDURE:

STEP 1:

Connect 4 pcs to one switch using copper straight-through cables.

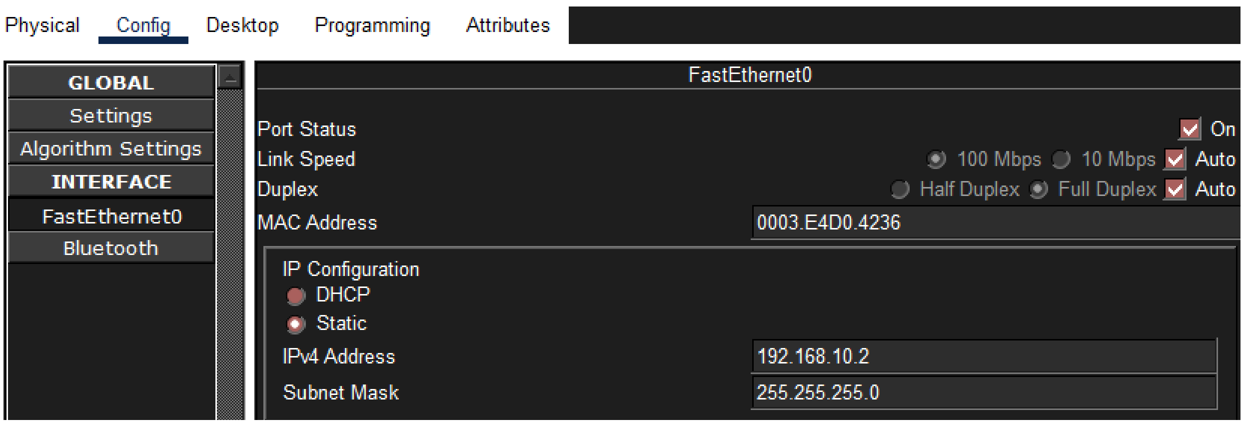


Step 2:

Assign IP Addresses and subnet masks to all the PCs. This will be done in such a way that two PCs will belong to VLAN 10 and two will belong to VLAN 20.

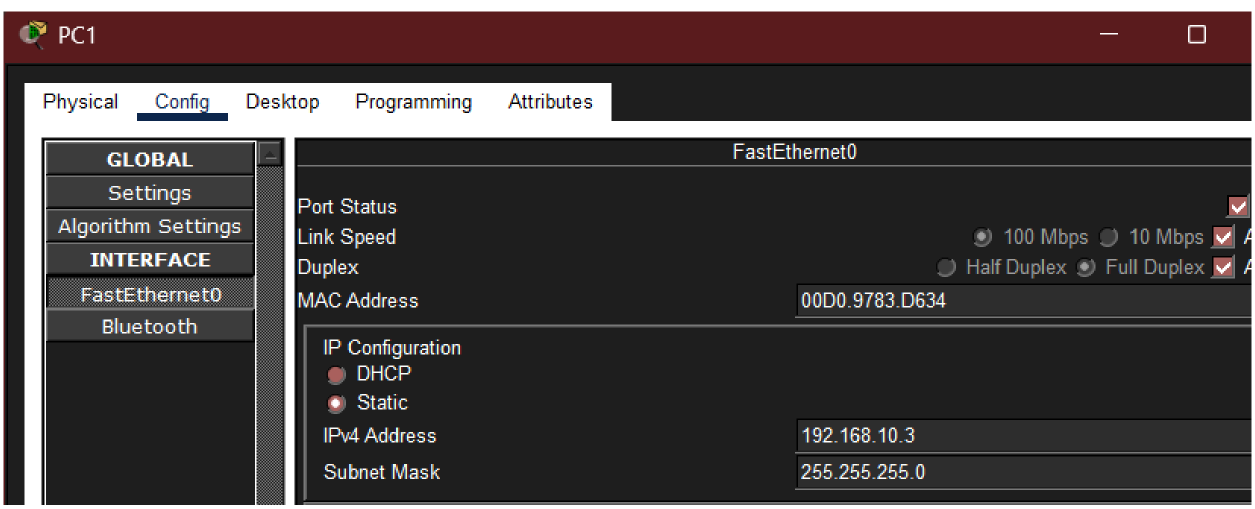
PC0

- IP Address: **192.168.10.2**
- Subnet Mask: **255.255.255.0**



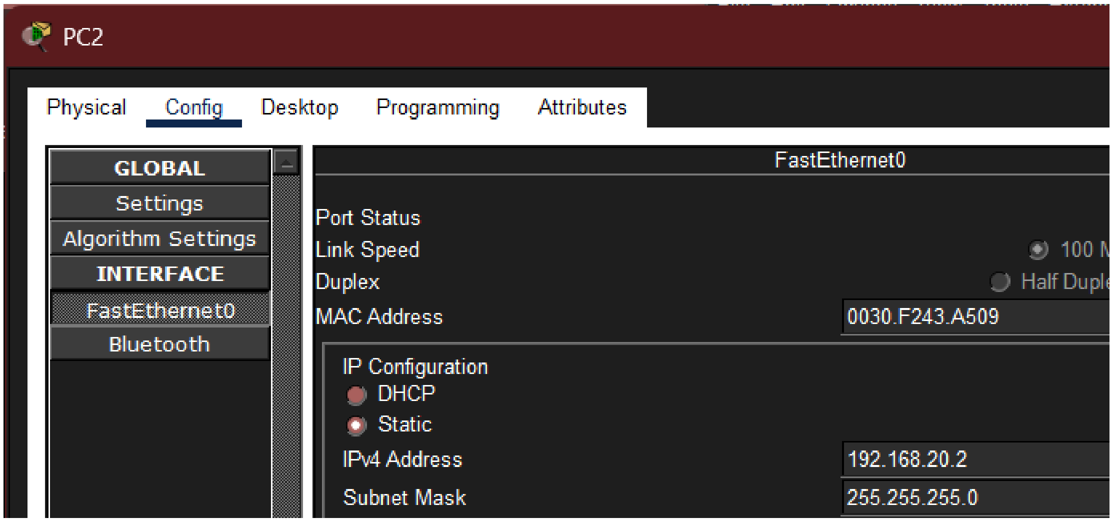
PC1

- IP Address: **192.168.10.3**
- Subnet Mask: **255.255.255.0**



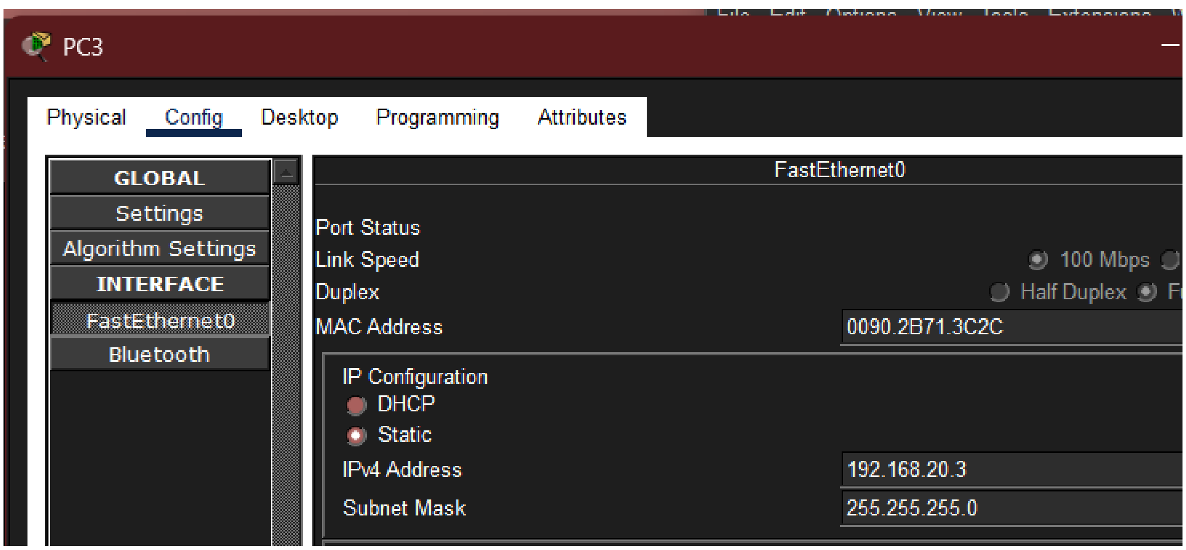
PC2

- IP Address: **192.168.20.2**
- Subnet Mask: **255.255.255.0**



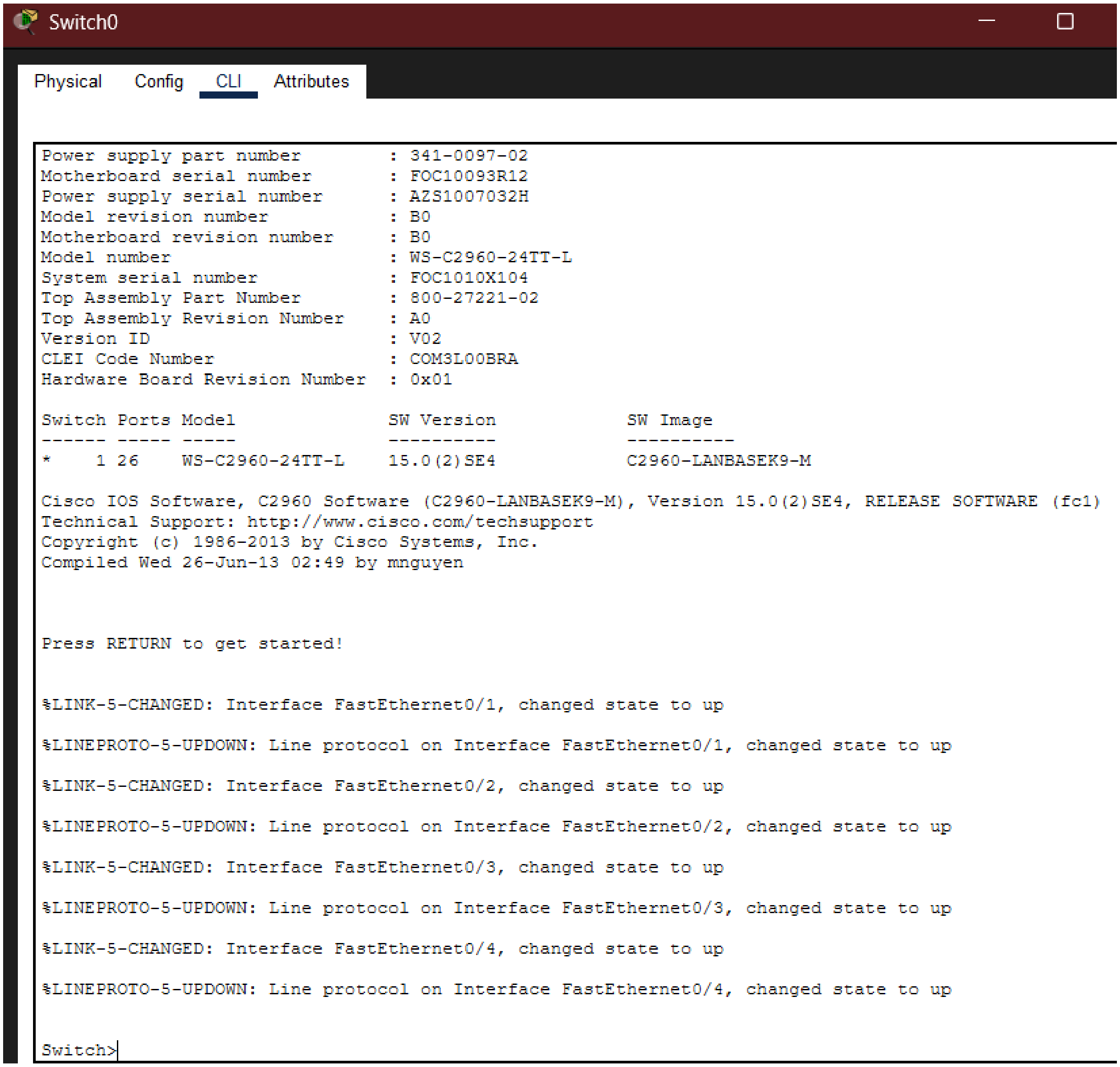
PC3

- IP Address: **192.168.20.3**
- Subnet Mask: **255.255.255.0**



Step 3:

Now we will access the Switch CLI.
Click on Switch and select the CLI tab and press enter to access the command prompt.



Switch0

Physical Config CLI Attributes

```
Power supply part number          : 341-0097-02
Motherboard serial number         : FOC10093R12
Power supply serial number        : AZS1007032H
Model revision number             : B0
Motherboard revision number       : B0
Model number                      : WS-C2960-24TT-L
System serial number              : FOC1010X104
Top Assembly Part Number          : 800-27221-02
Top Assembly Revision Number      : A0
Version ID                        : V02
CLEI Code Number                  : COM3L00BRA
Hardware Board Revision Number    : 0x01

Switch Ports Model                SW Version            SW Image
------ ----- -----                ----------            ----------
*    1 26    WS-C2960-24TT-L      15.0(2)SE4            C2960-LANBASEK9-M

Cisco IOS Software, C2960 Software (C2960-LANBASEK9-M), Version 15.0(2)SE4, RELEASE SOFTWARE (fc1)
Technical Support: http://www.cisco.com/techsupport
Copyright (c) 1986-2013 by Cisco Systems, Inc.
Compiled Wed 26-Jun-13 02:49 by mnguyen



Press RETURN to get started!


%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up

%LINK-5-CHANGED: Interface FastEthernet0/2, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to up

%LINK-5-CHANGED: Interface FastEthernet0/3, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/3, changed state to up

%LINK-5-CHANGED: Interface FastEthernet0/4, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/4, changed state to up


Switch>
```

Step 4:

We will enter a code that will create two VLANs A and B.

```
Switch> enable
Switch#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#vlan 10
Switch(config-vlan)#name A
Switch(config-vlan)#exit
Switch(config)#Vlan 20
Switch(config-vlan)#name B
Switch(config-vlan)#exit
Switch(config)#
```

Step 5:

Assign ports to VLAN 10 and configure ports Fa0/1 and Fa0/2

```
Switch(config-vlan)#exit
Switch(config)#interface range FastEthernet0/1-2
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 10
Switch(config-if-range)#exit
Switch(config)#
```

Step 6:

Assign ports to VLAN 20 and configure ports Fa0/3 and Fa0/4

```
Switch(config)#interface range FastEthernet0/3-4
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 20
Switch(config-if-range)#exit
```

Step 7:

Save the Switch configuration

```
Switch(config-if-range)#switchport access vlan 20
Switch(config-if-range)#exit
Switch(config)#end
Switch#
%SYS-5-CONFIG_I: Configured from console by console
copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
```

Step 8:

Verify VLAN configuration

```
[OK]
Switch#show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/5, Fa0/6, Fa0/7, Fa0/8
                                                Fa0/9, Fa0/10, Fa0/11, Fa0/12
                                                Fa0/13, Fa0/14, Fa0/15, Fa0/16
                                                Fa0/17, Fa0/18, Fa0/19, Fa0/20
                                                Fa0/21, Fa0/22, Fa0/23, Fa0/24
                                                Gig0/1, Gig0/2
10   A                                active    Fa0/1, Fa0/2
20   B                                active    Fa0/3, Fa0/4
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
Switch#
```

Step 9:

Now we will test communication between these PCs.

Scenario 1:

Communication within VLAN 10

We ping PC1 from PC0

```
C:\>ping 192.168.10.3

Pinging 192.168.10.3 with 32 bytes of data:

Reply from 192.168.10.3: bytes=32 time<1ms TTL=128
Reply from 192.168.10.3: bytes=32 time<1ms TTL=128
Reply from 192.168.10.3: bytes=32 time<1ms TTL=128
Reply from 192.168.10.3: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.10.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

Scenario 2:

Communication within VLAN 20

We ping PC3 from PC2

```
C:\>ping 192.168.20.3

Pinging 192.168.20.3 with 32 bytes of data:

Reply from 192.168.20.3: bytes=32 time<1ms TTL=128
Reply from 192.168.20.3: bytes=32 time<1ms TTL=128
Reply from 192.168.20.3: bytes=32 time<1ms TTL=128
Reply from 192.168.20.3: bytes=32 time=3ms TTL=128

Ping statistics for 192.168.20.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 3ms, Average = 0ms
```

Scenario 3:

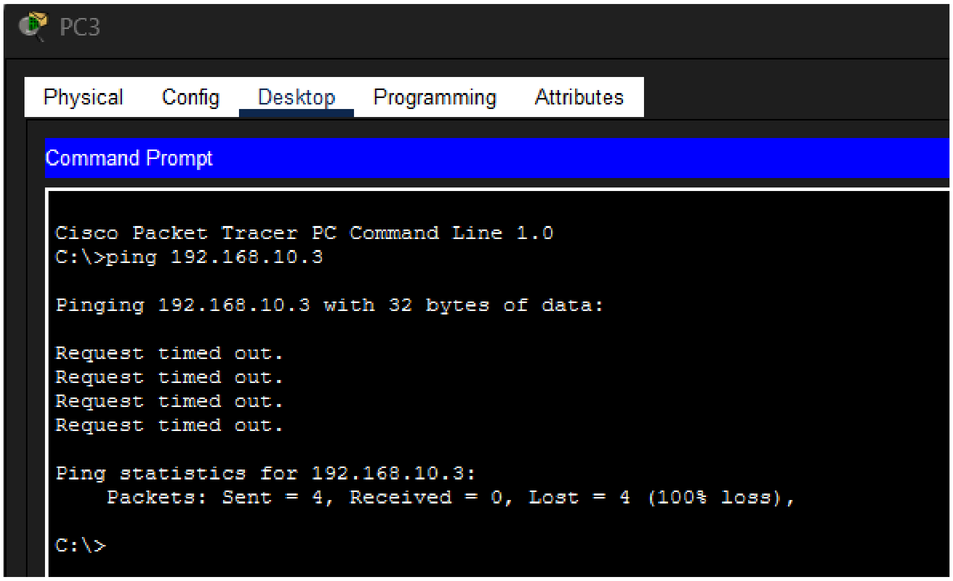
Communication between different VLAN

We ping PC2 from PC0

```
C:\>ping 192.168.20.2

Pinging 192.168.20.2 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.20.2:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

Scenario 4:
Communication between different VLAN

We ping PC1 from PC3



Observations

- Devices within the same VLAN communicated successfully.
- Devices belonging to different VLANs could not communicate.
- The switch isolated traffic between VLANs.
- Inter-VLAN communication requires a router or a Layer 3 switch.

Result

Two VLANs (**VLAN 10 – A** and **VLAN 20 – B**) were successfully created and configured on a single switch. Ports were assigned to their respective VLANs, and communication was verified. Devices within the same VLAN communicated successfully, while communication between different VLANs failed, demonstrating that a Layer 2 switch cannot route traffic between VLANs without a router or Layer 3 switch.

Lab # 07

Inter-VLAN Routing (Router-on-a-Stick)

Inter-VLAN Routing (Router-on-a-Stick):

Expand on the VLAN lab by adding a router and configuring a "router-on-a-stick" setup using subinterfaces to allow devices in different VLANs to communicate.

Objective

- To configure inter-VLAN routing using the Router-on-a-Stick method.
- To enable communication between different VLANs through a router.
- To configure router subinterfaces and a trunk link.

Equipment Required

- 1 Cisco Router
- 1 Cisco Switch (2960)
- 4 PCs
- UTP Straight-Through Cables
- Cisco Packet Tracer

Theory

Inter-VLAN Routing allows devices in different VLANs to communicate. In the **Router-on-a-Stick** method, a single physical router interface is divided into multiple **subinterfaces**, with each subinterface assigned to a different VLAN. The link between the router and the switch is configured as a **trunk** to carry traffic for multiple VLANs.

Procedure

1. Create VLANs on the switch.
2. Assign switch ports to the appropriate VLANs.
3. Configure the switch port connected to the router as a trunk.
4. Create router subinterfaces for each VLAN.
5. Assign IP addresses to the subinterfaces.
6. Configure the default gateway on each PC.
7. Verify communication using the **ping** command.

Observation

- Devices within the same VLAN communicated successfully.
- After configuring Router-on-a-Stick, devices in different VLANs were also able to communicate.

Result

Inter-VLAN routing was successfully configured using the Router-on-a-Stick method, allowing communication between different VLANs.

Precautions

- Configure the switch port connected to the router as a trunk.
- Use the correct VLAN IDs on router subinterfaces.
- Configure the correct default gateway on each PC.
- Save the configurations after completion.

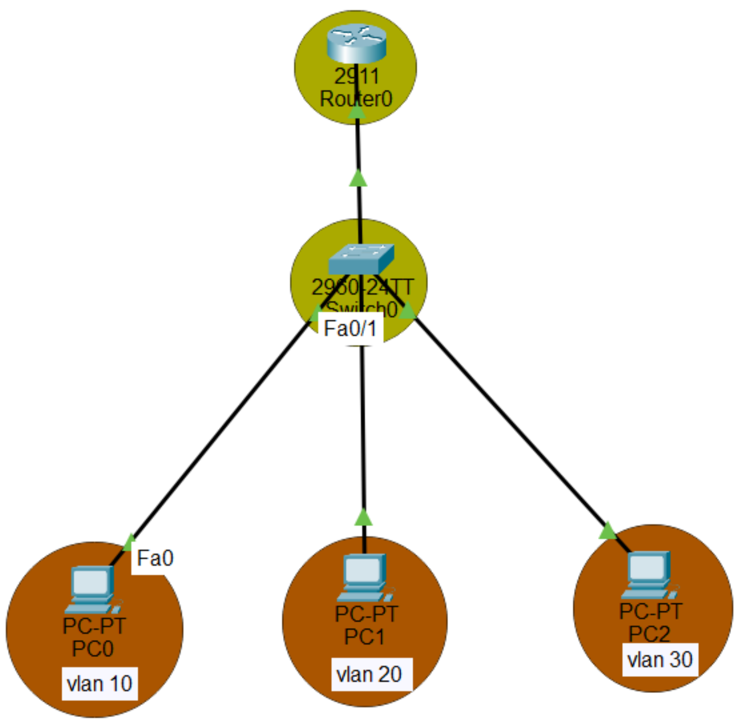


Figure: ***Topology***

```
Switch(config)#vlan 10
Switch(config-vlan)#name ABC
Switch(config-vlan)#exit
Switch(config)#interface f0/1
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 10
Switch(config-if)#exit
Switch(config)#do show vlan br

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/2, Fa0/3, Fa0/4, Fa0/5
                                                Fa0/6, Fa0/7, Fa0/8, Fa0/9
                                                Fa0/10, Fa0/11, Fa0/12, Fa0/13
                                                Fa0/14, Fa0/15, Fa0/16, Fa0/17
                                                Fa0/18, Fa0/19, Fa0/20, Fa0/21
                                                Fa0/22, Fa0/23, Fa0/24, Gig0/1
                                                Gig0/2
10   ABC                              active    Fa0/1
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
```

Figure: ***Vlan 10 Configuration***

```
Switch(config)#vlan 20
Switch(config-vlan)#name DEF
Switch(config-vlan)#interface f0/2
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 20
Switch(config-if)#exit
Switch(config)#do show vlan br

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/3, Fa0/4, Fa0/5, Fa0/6
                                                Fa0/7, Fa0/8, Fa0/9, Fa0/10
                                                Fa0/11, Fa0/12, Fa0/13, Fa0/14
                                                Fa0/15, Fa0/16, Fa0/17, Fa0/18
                                                Fa0/19, Fa0/20, Fa0/21, Fa0/22
                                                Fa0/23, Fa0/24, Gig0/1, Gig0/2
10   ABC                              active    Fa0/1
20   DEF                              active    Fa0/2
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
```

Figure: ***Vlan 20 Configuration***

```
Switch(config)#vlan 30
Switch(config-vlan)#name GHI
Switch(config-vlan)#exit
Switch(config)#interface f0/3
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 30
Switch(config-if)#exit
Switch(config)#do show vlan br

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/4, Fa0/5, Fa0/6, Fa0/7
                                                Fa0/8, Fa0/9, Fa0/10, Fa0/11
                                                Fa0/12, Fa0/13, Fa0/14, Fa0/15
                                                Fa0/16, Fa0/17, Fa0/18, Fa0/19
                                                Fa0/20, Fa0/21, Fa0/22, Fa0/23
                                                Fa0/24, Gig0/1, Gig0/2
10   ABC                              active    Fa0/1
20   DEF                              active    Fa0/2
30   GHI                              active    Fa0/3
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
```

Figure: ***Vlan 30 Configuration***

```
Router(config)#inter
Router(config)#interface g0/0.10
Router(config-subif)#
%LINK-3-UPDOWN: Interface GigabitEthernet0/0.10, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.10, changed state to up

Router(config-subif)#encapsulation dot1Q 10
Router(config-subif)#ip address 192.168.10.1 255.255.255.0
Router(config-subif)#exit
Router(config)#interface g0/0.20
Router(config-subif)#
%LINK-3-UPDOWN: Interface GigabitEthernet0/0.20, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.20, changed state to up

Router(config-subif)#encapsulation dot1
Router(config-subif)#encapsulation dot1Q 20
Router(config-subif)#ip add
Router(config-subif)#ip address 192.168.20.1 255.255.255.0
Router(config-subif)#exit
Router(config)#
Router(config)#interface g0/0.30
Router(config-subif)#
%LINK-3-UPDOWN: Interface GigabitEthernet0/0.30, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.30, changed state to up

Router(config-subif)#encapsulation dot
Router(config-subif)#encapsulation dot1Q 30
Router(config-subif)#ip address 192.168.30.1 255.255.255.0
Router(config-subif)#exit
Router(config)#
```

Figure: ***Configuration f Sub interfaces***

```
VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/4, Fa0/5, Fa0/6, Fa0/7
                                                Fa0/8, Fa0/9, Fa0/10, Fa0/11
                                                Fa0/12, Fa0/13, Fa0/14, Fa0/15
                                                Fa0/16, Fa0/17, Fa0/18, Fa0/19
                                                Fa0/20, Fa0/21, Fa0/22, Fa0/23
                                                Gig0/1, Gig0/2
10   ABC                              active    Fa0/1
20   DEF                              active    Fa0/2
30   GHI                              active    Fa0/3
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
Switch(config)#do show interface trunk
Port        Mode         Encapsulation  Status        Native vlan
Fa0/24      on           802.1q         trunking      1

Port        Vlans allowed on trunk
Fa0/24      1-1005

Port        Vlans allowed and active in management domain
Fa0/24      1,10,20,30

Port        Vlans in spanning tree forwarding state and not pruned
Fa0/24      1,10,20,30
```

Figure: Verification of Subinterfaces

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.20.2

Pinging 192.168.20.2 with 32 bytes of data:

Reply from 192.168.20.2: bytes=32 time=1ms TTL=127
Reply from 192.168.20.2: bytes=32 time<1ms TTL=127
Reply from 192.168.20.2: bytes=32 time<1ms TTL=127
Reply from 192.168.20.2: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.20.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

Ping status

Lab # 08

Basic Wireless Network Setup Using Cisco Packet Tracer

Description of Lab:

Part 1: IP Address Assignment and Subnetting:

Objective:

The objective of this lab is to learn IPv4 subnetting by dividing the network 192.168.100.0/24 into two subnets. After subnetting, the network address, usable IP address range, and broadcast address for each subnet are identified. The PCs are then assigned appropriate IP addresses and tested using the ping command in Cisco Packet Tracer.

Part 2: Creating Subnets from a Class C Address Using VLANs

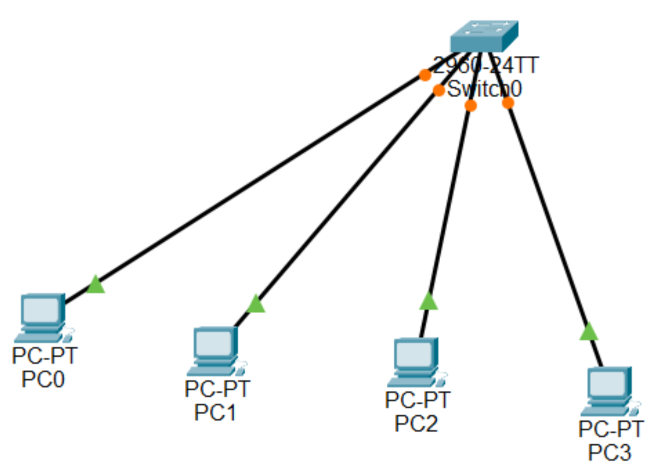
Objective:

The objective of this lab is to divide the Class C network 192.168.100.0/24 into four subnets for different departments (HR, Sales, IT, and Marketing). VLANs are created on the switch, switch ports are assigned to the appropriate VLANs, IP addresses are configured on the PCs, and connectivity is tested in Cisco Packet Tracer.

Screenshot from Packet Tracer Workspace:

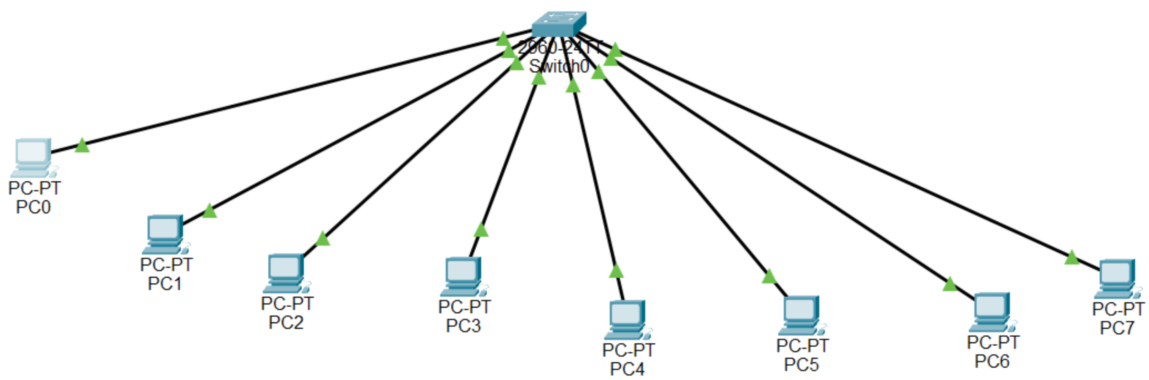
Part 1

i. 1 Switch
ii. 4 PCs
iii. All cable connections



Part 2

- 1 Switch
- 8 PCs
- VLAN topology
- Cable connections



Configuration Steps:

Part 1 two subnets

Step 1

Open Cisco Packet Tracer.

Step 2

Place one **2960 Switch** and four PCs.

Step 3

Connect all PCs to the switch using **Copper Straight-Through Cable**.

Step 4

Subnet the network.

Network Address

192.168.100.0/24

Need 2 subnets

New subnet mask

255.255.255.128 (/25)

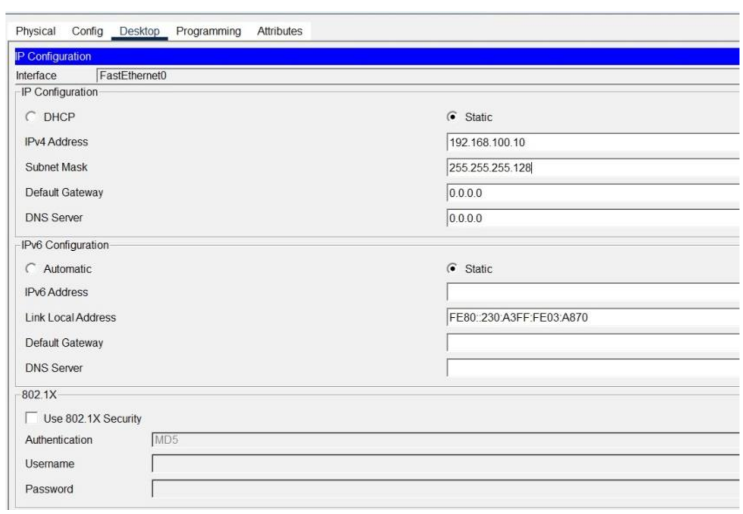
Subnet Table

Subnet	Network Address	Usable Range	Broadcast
Subnet 1	192.168.100.0	192.168.100.1 – 192.168.100.126	192.168.100.127
Subnet 2	192.168.100.128	192.168.100.129 – 192.168.100.254	192.168.100.255

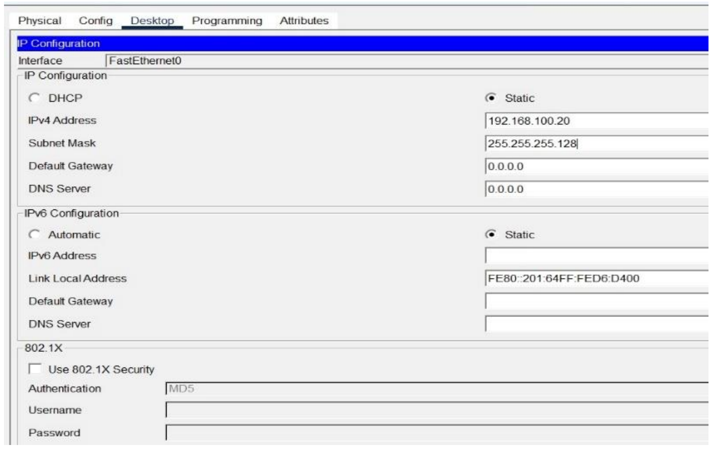
Step 5

Assign IP addresses.

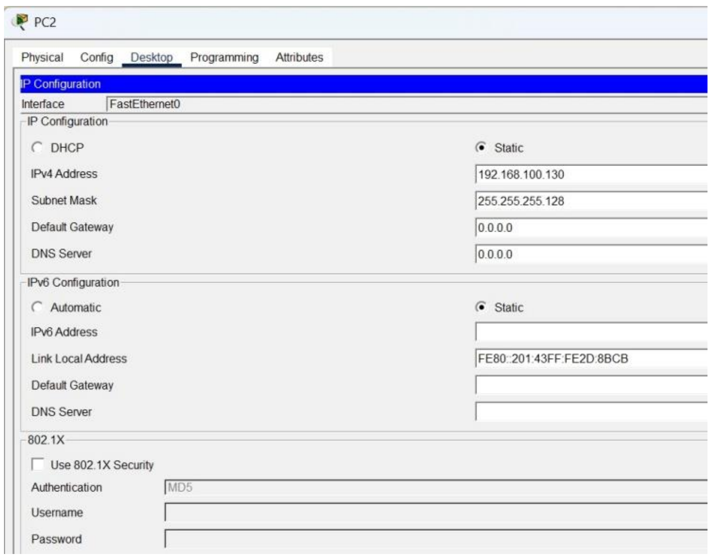
PC	IP Address	Subnet Mask
PC0	192.168.100.10	255.255.255.128
PC1	192.168.100.20	255.255.255.128
PC2	192.168.100.130	255.255.255.128
PC3	192.168.100.140	255.255.255.128



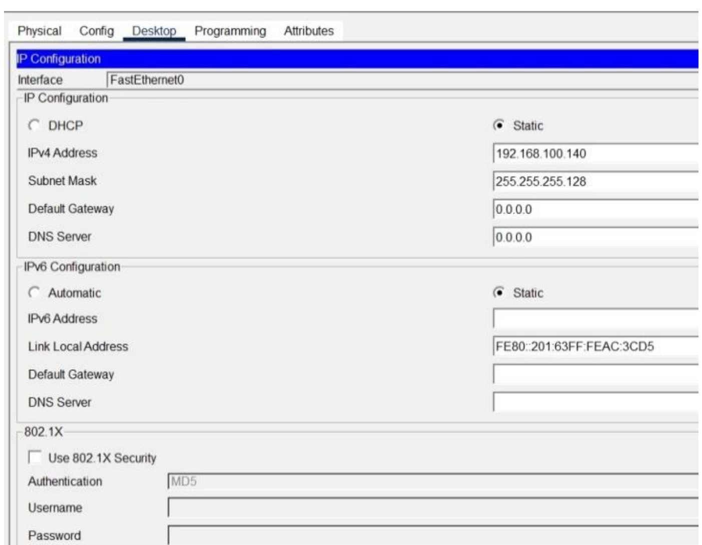
Pc 0



Pc 1



Pc 2

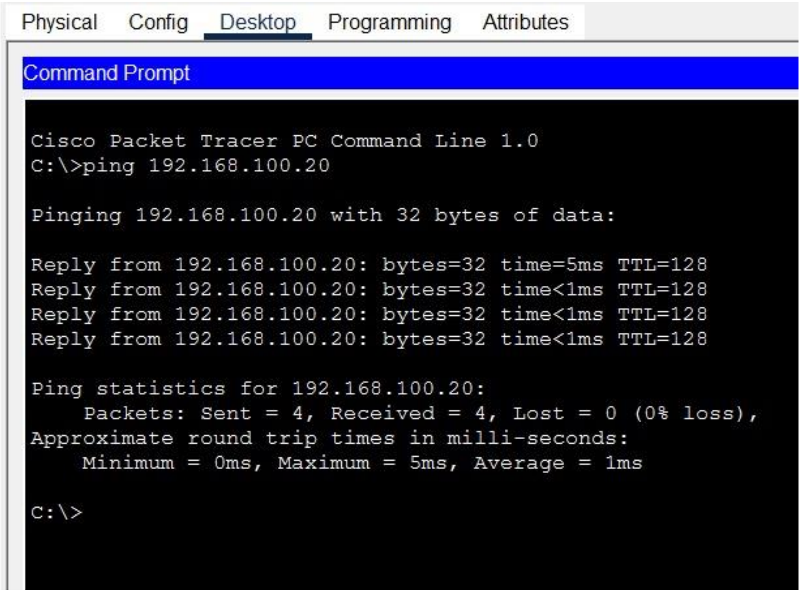


Step 6

Test the network using

Pc 3

ping 192.168.100.20



Part 2 – Four Departments Using VLANs

Step 1

Open a **new Packet Tracer workspace**.

Step 2

Place

- One 2960 Switch
- Eight PCs

Step 3

Connect every PC to the switch.

Step 4

Subnet the network.

Network

192.168.100.0/24
Need
4 Subnets

New subnet mask

255.255.255.192 (/26)

Subnet Table

Department	Network	Usable Range	Broadcast
HR	192.168.100.0/26	192.168.100.1–62	192.168.100.63
Sales	192.168.100.64/26	192.168.100.65–126	192.168.100.127
IT	192.168.100.128/26	192.168.100.129–190	192.168.100.191
Marketing	192.168.100.192/26	192.168.100.193–254	192.168.100.255

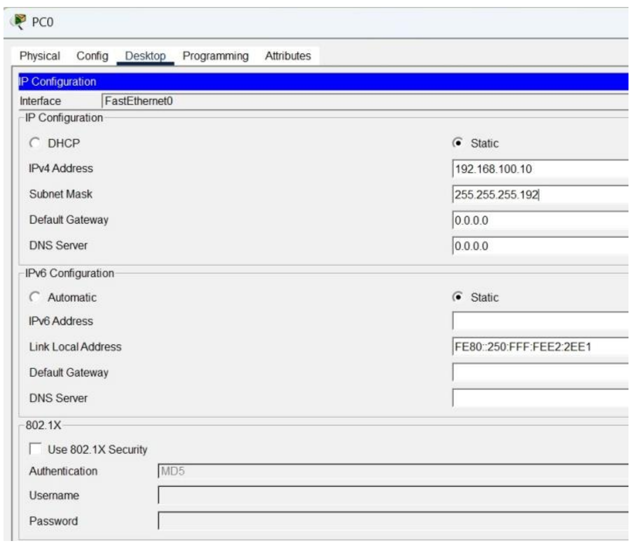
Step 5

Assign IP addresses.

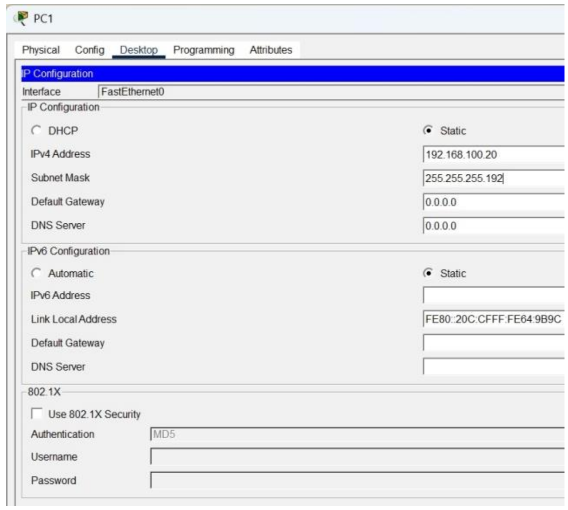
Department	PC	IP Address
HR	PC0	192.168.100.10
HR	PC1	192.168.100.20
Sales	PC2	192.168.100.70
Sales	PC3	192.168.100.80
IT	PC4	192.168.100.130
IT	PC5	192.168.100.140
Marketing	PC6	192.168.100.200
Marketing	PC7	192.168.100.210

Subnet Mask for all:

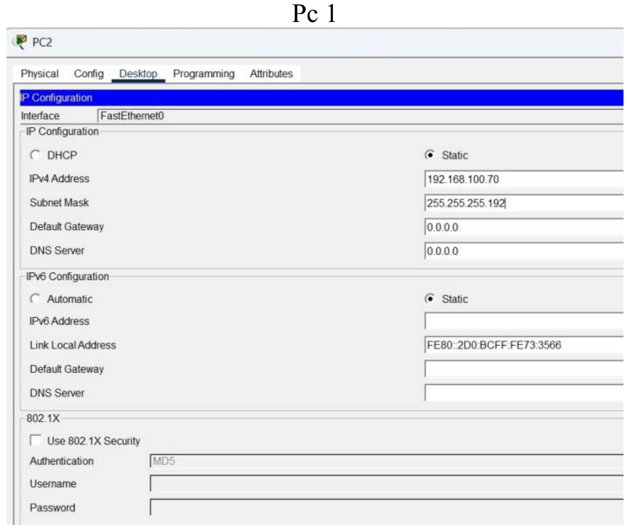
255.255.255.192



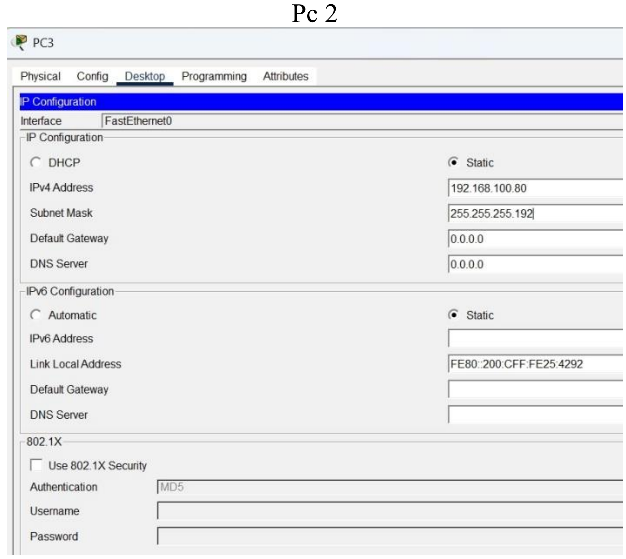
Pc 0


PC1
Physical
Config
Desktop
Programming
Attributes
IP Configuration
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
192.168.100.20
Subnet Mask
255.255.255.192
Default Gateway
0.0.0.0
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
Link Local Address
FE80::20C:CFFF:FE64:9B9C
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password

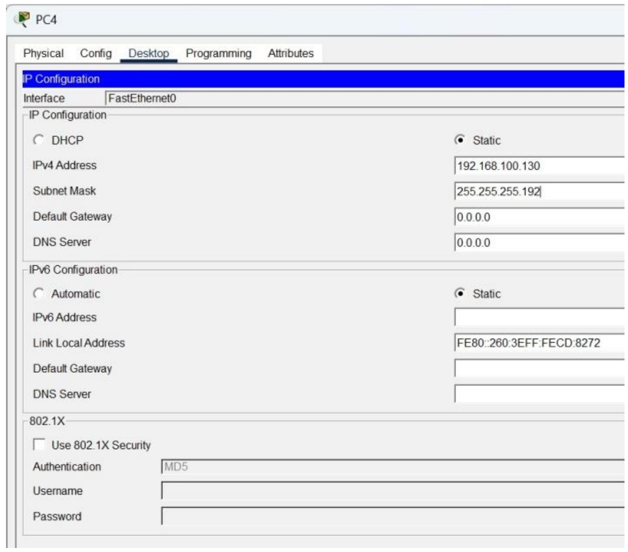
Pc 1


PC2
Physical
Config
Desktop
Programming
Attributes
IP Configuration
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
192.168.100.70
Subnet Mask
255.255.255.192
Default Gateway
0.0.0.0
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
Link Local Address
FE80::2D0:BCFF:FE73:3566
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password

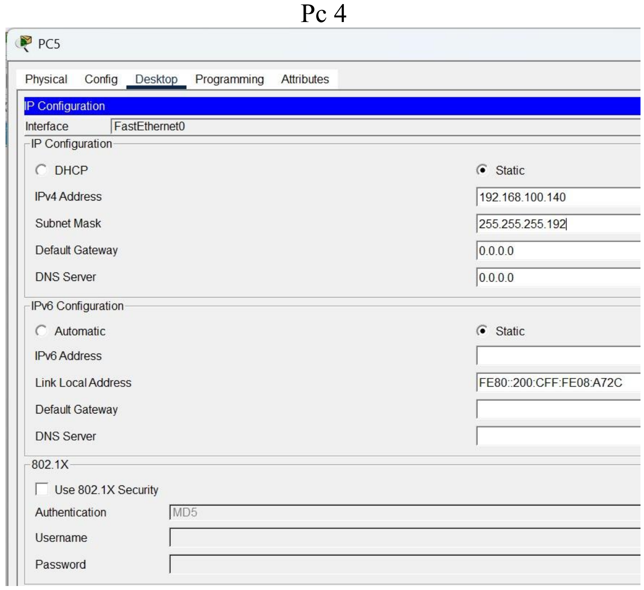
Pc 2


PC3
Physical
Config
Desktop
Programming
Attributes
IP Configuration
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
192.168.100.80
Subnet Mask
255.255.255.192
Default Gateway
0.0.0.0
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
Link Local Address
FE80::200:CFF:FE25:4292
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password

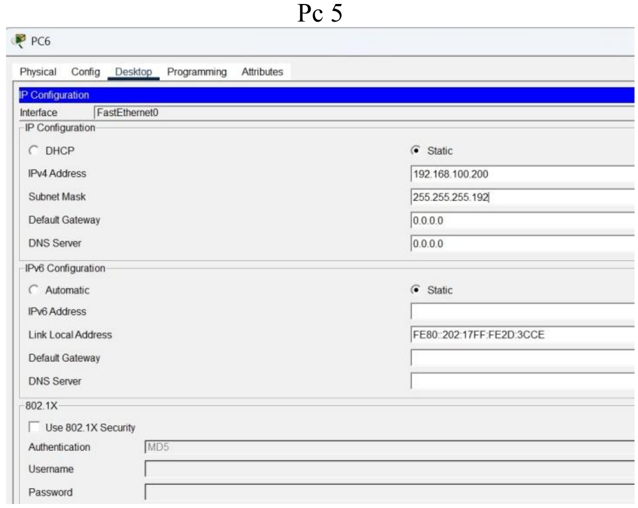
Pc 3


PC4
Physical
Config
Desktop
Programming
Attributes
IP Configuration
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
192.168.100.130
Subnet Mask
255.255.255.192
Default Gateway
0.0.0.0
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
Link Local Address
FE80::260:3EFF:FECD:8272
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password

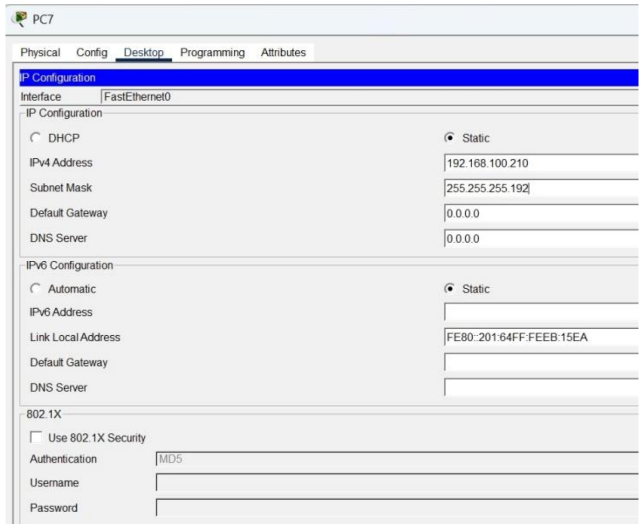
Pc 4


PC5
Physical
Config
Desktop
Programming
Attributes
IP Configuration
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
192.168.100.140
Subnet Mask
255.255.255.192
Default Gateway
0.0.0.0
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
Link Local Address
FE80::200:CFF:FE08:A72C
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password

Pc 5


PC6
Physical
Config
Desktop
Programming
Attributes
IP Configuration
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
192.168.100.200
Subnet Mask
255.255.255.192
Default Gateway
0.0.0.0
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
Link Local Address
FE80::202:17FF:FE2D:3CCE
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password

Pc 6



Pc 7

CLI Commands


```
Switch>enable
Switch#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#vlan 10
Switch(config-vlan)#name HR
Switch(config-vlan)#exit
Switch(config)#vlan 20
Switch(config-vlan)#name Sales
Switch(config-vlan)#exit
Switch(config)#vlan 30
Switch(config-vlan)#name IT
Switch(config-vlan)#exit
Switch(config)#vlan 40
Switch(config-vlan)#name Marketing
Switch(config-vlan)#exit
Switch(config)#interface range fa0/1-2
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 10
Switch(config-if-range)#exit
Switch(config)#interface range fa0/3-4
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 20
Switch(config-if-range)#exit
Switch(config)#interface range fa0/5-6
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 30
Switch(config-if-range)#exit
Switch(config)#interface range fa0/7-8
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 40
Switch(config-if-range)#exit
Switch(config)#end
Switch#
%SYS-5-CONFIG_I: Configured from console by console

Switch#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
```

Verification command

```
Switch#show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/9, Fa0/10, Fa0/11, Fa0/12
                                                Fa0/13, Fa0/14, Fa0/15, Fa0/16
                                                Fa0/17, Fa0/18, Fa0/19, Fa0/20
                                                Fa0/21, Fa0/22, Fa0/23, Fa0/24
                                                Gig0/1, Gig0/2
10   HR                               active    Fa0/1, Fa0/2
20   Sales                            active    Fa0/3, Fa0/4
30   IT                               active    Fa0/5, Fa0/6
40   Marketing                        active    Fa0/7, Fa0/8
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
Switch#
```

Lab # 09

Configuring DNS & Web Server

Description of Lab:

This lab demonstrates how to configure a server to provide two essential network services — **DNS (Domain Name System)** and **HTTP (Web Server)** — within a switch-based LAN using Cisco Packet Tracer. The DNS service resolves a human-readable domain name (e.g., www.astrolab.com) into the server's IP address, while the HTTP service hosts a webpage that client PCs can access using that domain name instead of the raw IP address. The lab also required correcting an IP subnet mismatch between the server and the client PCs before DNS resolution could succeed.

Devices used: 1 Server-PT, 1 Cisco 2960 Switch, 3 PC-PT end devices, Copper Straight-Through cables.

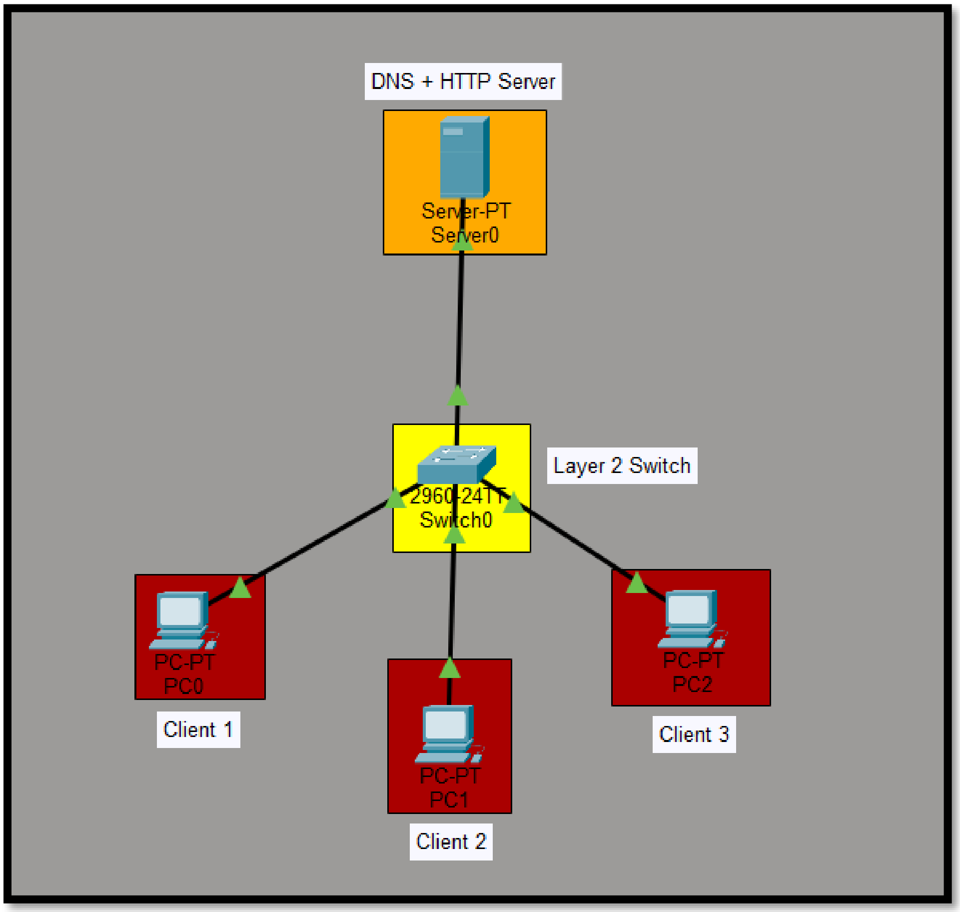


Figure 2:-General Topology

Configuration Steps:

- Placed one Server, one Switch, and three PCs in the workspace.
- Connected all devices to the switch using Copper Straight-Through cables.
- Assigned static IP addresses to the Server and all PCs on the `192.168.10.0/24` subnet.
- Set the DNS server address on each PC to point to `192.168.10.1` (the server).
- Enabled the DNS service on the server and added an A Record mapping `www.astrolab.com`→ `192.168.10.1`.
- Enabled the HTTP service on the server and customized the default webpage.
- Verified Layer 3 connectivity using `ping`.
- Identified and corrected a subnet mismatch — the server was initially configured on `192.168.11.1` while the PCs and DNS server field pointed to `192.168.10.1`, causing DNS request timeouts.
- Re-verified DNS resolution using `nslookup` after the fix.
- Accessed the website from client PCs using the domain name in the Web Browser application.

IP Addressing Table:

Device Name	Device Type	IP Address	Subnet Mask	Default Gateway	DNS Server
Server0	Server (DNS + HTTP)	192.168.10.1	255.255.255.0	0.0.0.0	192.168.10.1
PC0	Client PC	192.168.10.2	255.255.255.0	0.0.0.0	192.168.10.1
PC1	Client PC	192.168.10.3	255.255.255.0	0.0.0.0	192.168.10.1
PC2	Client PC	192.168.10.4	255.255.255.0	0.0.0.0	192.168.10.1

Figure 3:-Table For IP Addressing.

CLI Commands:

This lab primarily used GUI-based server/PC configuration rather than router or switch CLI. Commands used were from the PC Command Prompt:

ping 192.168.10.1
nslookup www.astrolab.com

Optional Switch CLI Documentation:

Switch> enable
Switch# configure terminal
Switch(config)# hostname SW-Lab9
SW-Lab9(config)# interface fastEthernet 0/1
SW-Lab9(config-if)# description Link-to-Server
SW-Lab9(config-if)# exit

Simulation Results:

Figure 4:-Server IP Configuration.

Figure 5:-PC0 IP Configuration(Example).

No.	Name	Type	Detail
0	www.astrolab.com	A Record	192.168.10.1

Figure 6:- DNS Services page Showing 'The A Record' Added.

HTTP: On / Off — HTTPS: On / Off

File Manager

	File Name	Edit	Delete
1	copyrights.html	(edit)	(delete)
2	cscoptlogo177x111.jpg		(delete)
3	helloworld.html	(edit)	(delete)
4	image.html	(edit)	(delete)
5	index.html	(edit)	(delete)

Figure 7:- HTTP Services Page (Showing Service Is ON).

```
C:\>ping 192.168.10.1

Pinging 192.168.10.1 with 32 bytes of data:

Reply from 192.168.10.1: bytes=32 time<1ms TTL=128
Reply from 192.168.10.1: bytes=32 time=6ms TTL=128
Reply from 192.168.10.1: bytes=32 time<1ms TTL=128
Reply from 192.168.10.1: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.10.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 6ms, Average = 1ms
```

Figure 8:-Successful Ping From PC0.

```
C:\>nslookup www.astrolab.com

Server: [192.168.10.1]
Address:  192.168.10.1

Non-authoritative answer:
Name:   www.astrolab.com
Address:   192.168.10.1
```

Figure 9:-Successful nslookup From PC0.

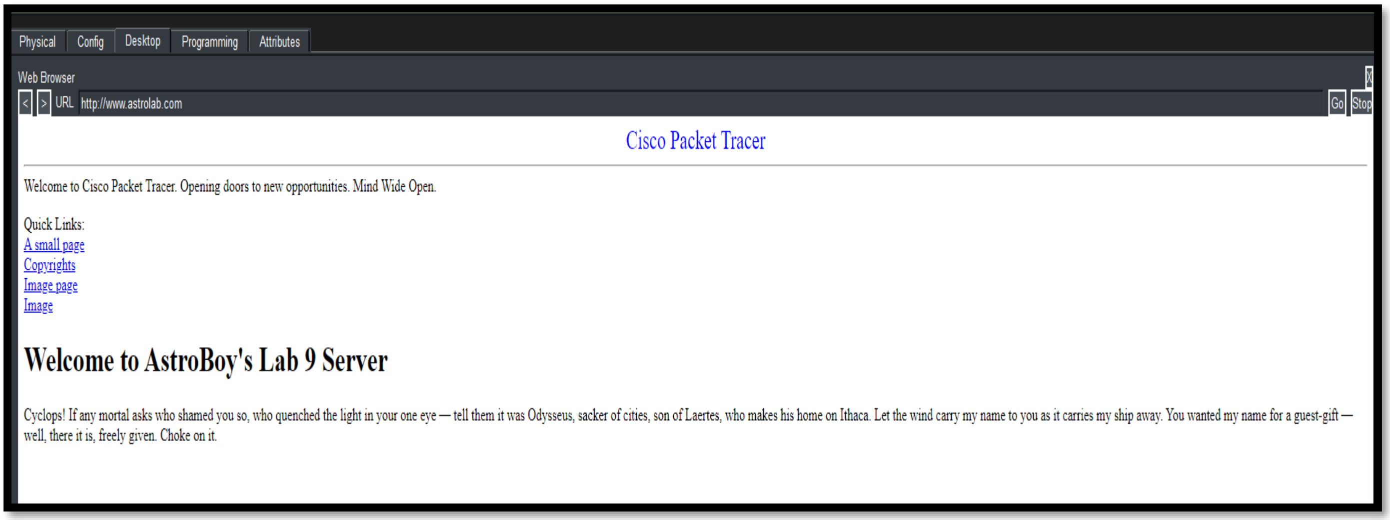


Figure 10:-Successful Web Browser Access.

```
Switch>enable
Switch#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#hostname SW-lab9
SW-lab9(config)#interface fastEthernet 0/1
SW-lab9(config-if)#description Link-to-Server
SW-lab9(config-if)#exit
SW-lab9(config)#exit
SW-lab9#
%SYS-5-CONFIG_I: Configured from console by console
show running-config
Building configuration...

Current configuration : 1109 bytes
!
version 15.0
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
!
hostname SW-lab9
!
!
!
!
!
!
spanning-tree mode pvst
spanning-tree extend system-id
!
interface FastEthernet0/1
 description Link-to-Server
!
```

Figure 11:-Switch CLI Configuration.

Conclusion:

The lab successfully demonstrates that a single server can provide both DNS and HTTP services simultaneously, allowing client devices to resolve domain names and access web content without needing to know the server's raw IP address — replicating how real-world domain-based browsing works.

Lab # 10

Basic Wireless Network Setup Using Cisco Packet Tracer

Lab Task Assigned

Basic Wireless Network Setup: Configure a wireless router to provide Wi-Fi access to wireless end devices (laptops, smartphones). Set up basic security like WPA2-PSK.

Lab Objectives

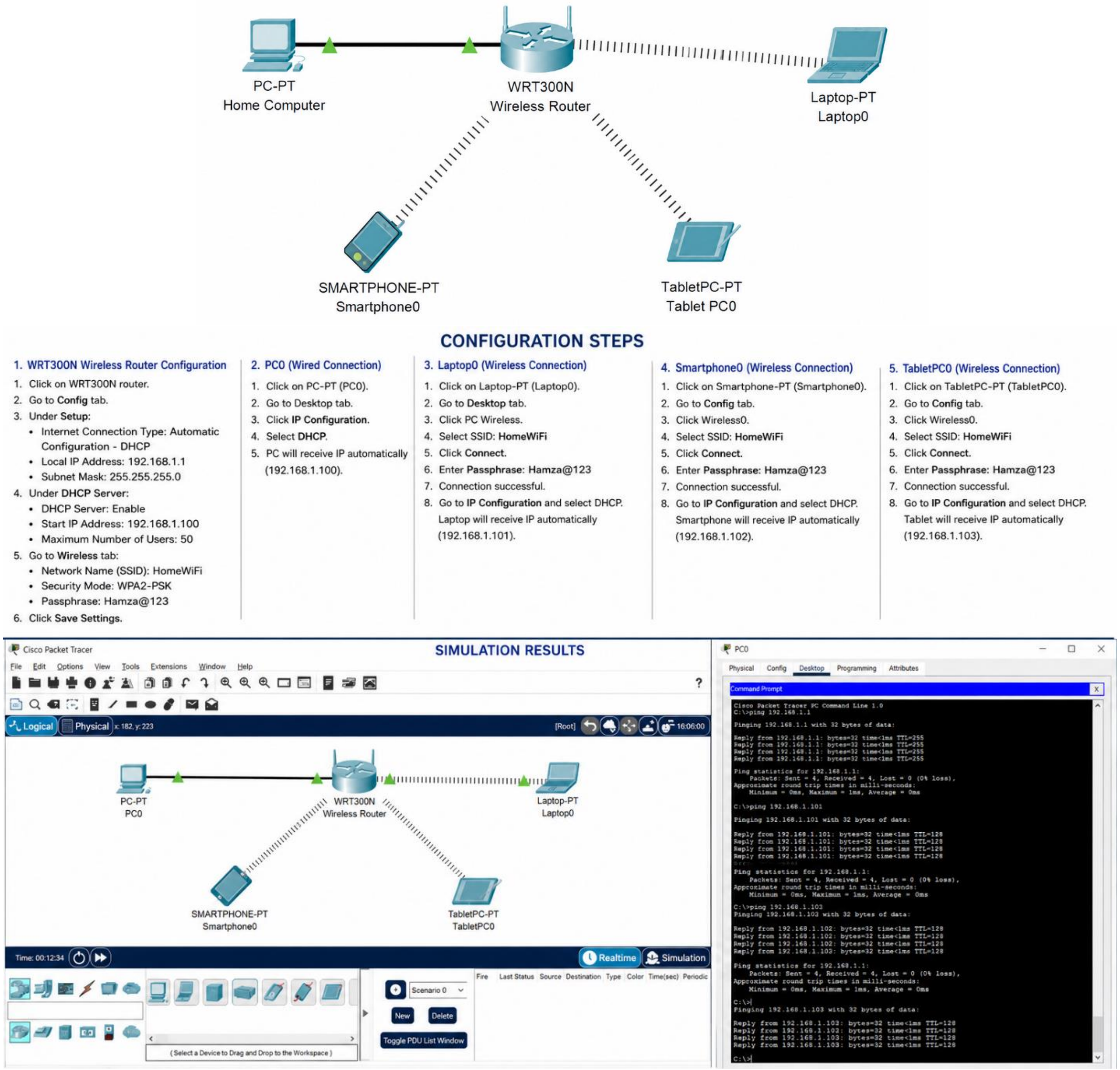
After completing this lab, students will be able to:

- Configure a wireless router in Cisco Packet Tracer.
- Provide Wi-Fi connectivity to wireless end devices.
- Configure a wireless network using **WPA2-PSK** security.
- Connect wired and wireless devices to the same network.
- Verify network connectivity using the **Ping** command.
- Test communication between all connected devices.

Components Required (Cisco Packet Tracer Software)

Components Used	Quantity
PC-PT (Home Computer)	1
WRT300N Wireless Router	1
Laptop-PT	1
Smartphone-PT	1
TabletPC-PT	1
Copper Straight-Through Cable	1

Cisco Design



Network Configuration Steps

Step 1: Place the Devices

- Add one **WRT300N Wireless Router**.
- Add one **PC-PT**.
- Add one **Laptop-PT**.
- Add one **Smartphone-PT**.

- Add one **TabletPC-PT**.

Step 2: Connect the Wired Device

- Connect the **PC-PT** to the **LAN port** of the WRT300N using a **Copper Straight-Through Cable**.

Step 3: Configure the Wireless Router

Open the **WRT300N Wireless Router**.

Basic Setup

- Internet Connection Type: **Automatic Configuration (DHCP)**
- Local IP Address: **192.168.1.1**
- Subnet Mask: **255.255.255.0**
- DHCP Server: **Enabled**
- Starting IP Address: **192.168.1.100**
- Maximum Number of Users: **50**

Step 4: Configure Wireless Settings

Open the **Wireless** tab.

Configure:

- Network Name (SSID): **HomeWiFi**
- Wireless Mode: **Mixed**
- SSID Broadcast: **Enabled**

Save the settings.

Step 5: Configure Wireless Security

Open **Wireless Security**.

Configure:

- Security Mode: **WPA2-Personal (WPA2-PSK)**
- Encryption: **AES**
- Passphrase: **Danial@123**

Save the configuration.

Step 6: Connect the Laptop

- Open **Laptop-PT**.
- Go to **Desktop → PC Wireless**.
- Select **HomeWiFi**.
- Click **Connect**.
- Enter the password:

Danial@123

- Obtain an IP address automatically using **DHCP**.

Step 7: Connect the Smartphone

- Open **Smartphone-PT**.
- Enable the wireless interface.
- Select **HomeWiFi**.
- Enter the password:

Danial@123

- Obtain an IP address automatically using **DHCP**.

Step 8: Connect the Tablet

- Open **TabletPC-PT**.
- Enable the wireless interface.
- Select **HomeWiFi**.
- Enter the password:

Danial@123

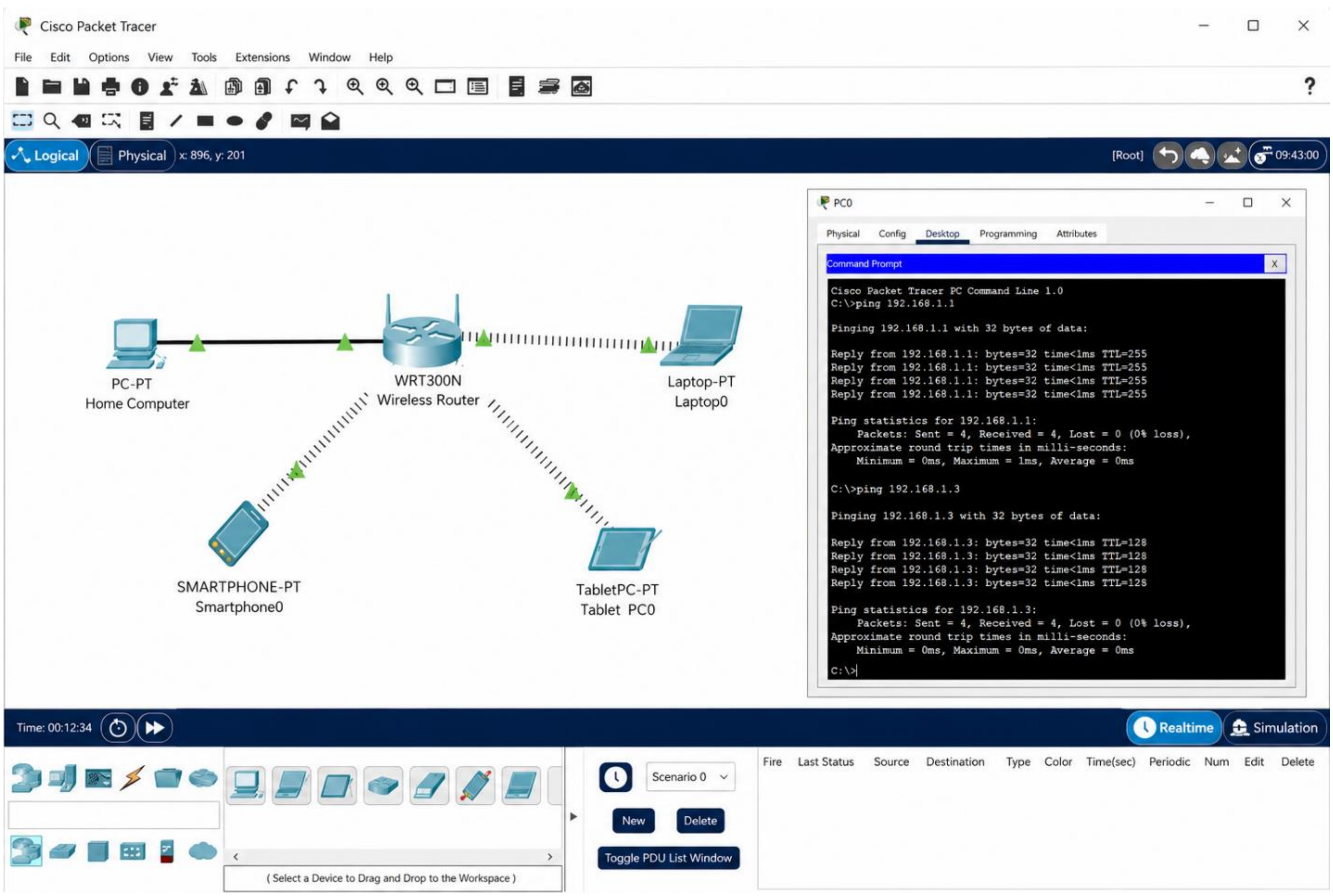
- Obtain an IP address automatically using **DHCP**.

Step 9: Verify Connectivity

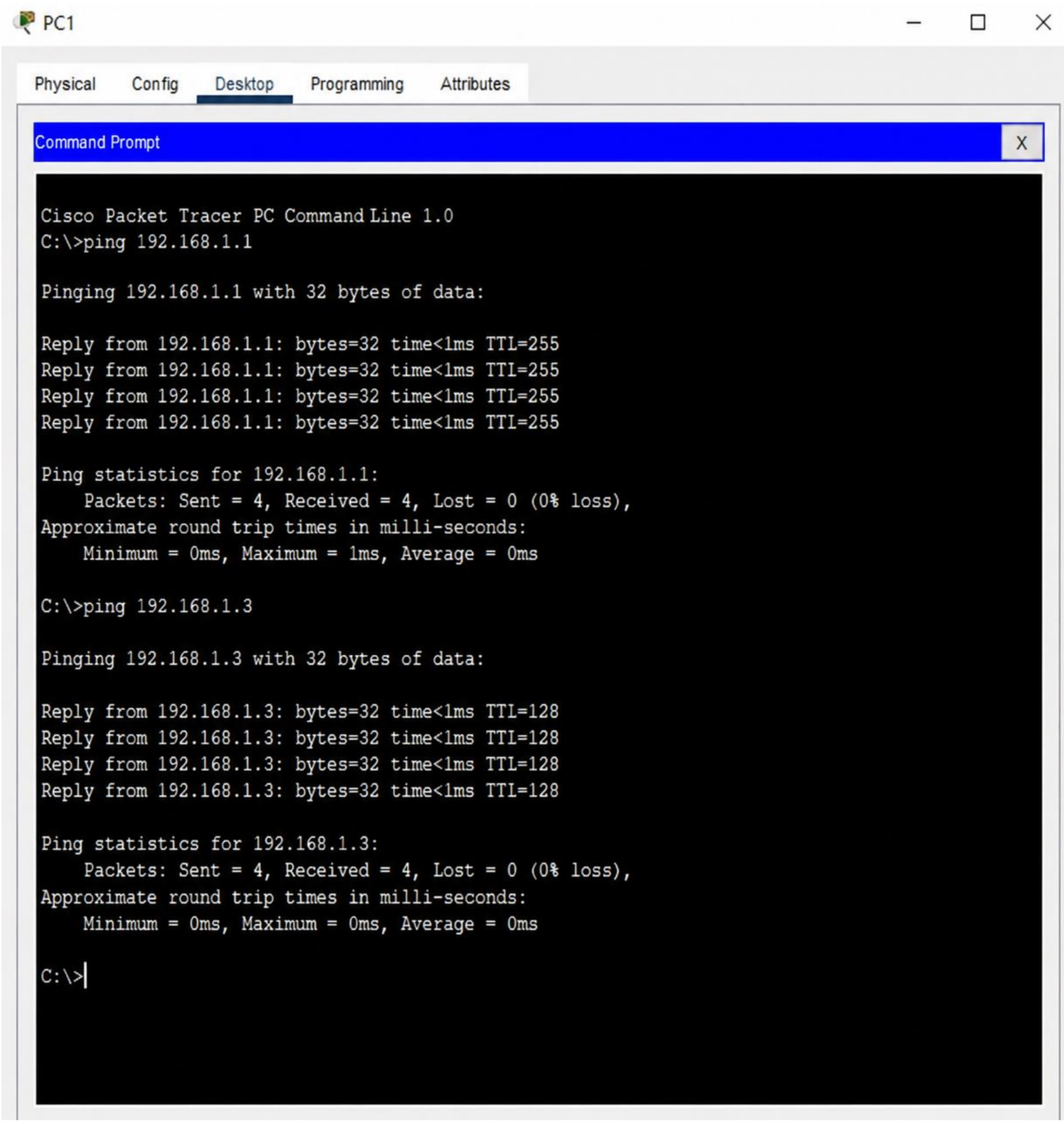
Open the **Command Prompt** on the PC.
Execute:

ping 192.168.1.1 ping 192.168.1.2
ping 192.168.1.3 ping 192.168.1.4

Simulation Results



PICTURE 12




```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.1

Pinging 192.168.1.1 with 32 bytes of data:

Reply from 192.168.1.1: bytes=32 time<1ms TTL=255
Reply from 192.168.1.1: bytes=32 time<1ms TTL=255
Reply from 192.168.1.1: bytes=32 time<1ms TTL=255
Reply from 192.168.1.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.1.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>ping 192.168.1.3

Pinging 192.168.1.3 with 32 bytes of data:

Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

PICTURE 2

Conclusion

- The wireless router successfully provides Wi-Fi connectivity.
- The laptop, smartphone, and tablet connect using **WPA2-PSK** authentication.
- All devices receive IP addresses through DHCP.
- The PC successfully pings the router and all wireless devices with **0% packet loss**, confirming a correctly configured and secure wireless network.